

Indian Summer Monsoon

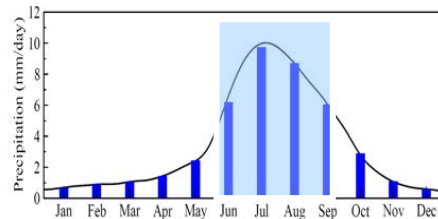
- ☐ Definition of monsoon
- ☐ Driving forces of the monsoon
- ☐ Synoptic features associated with onset & withdrawal
- ☐ Active and Break cycles of monsoon
- ☐ Rainfall distribution and rain bearing systems



What is a Monsoon?

Monsoon ~ 'mausim' → SEASON

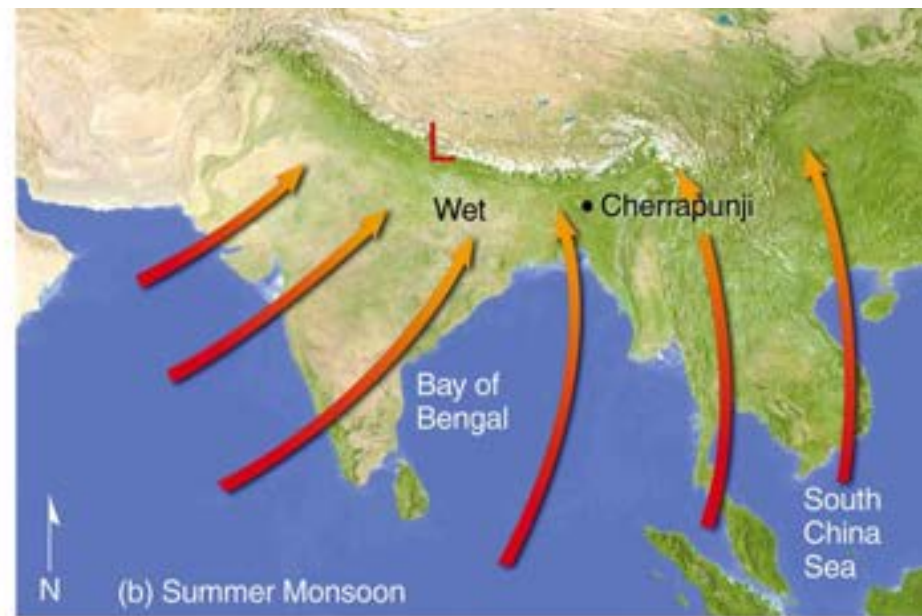
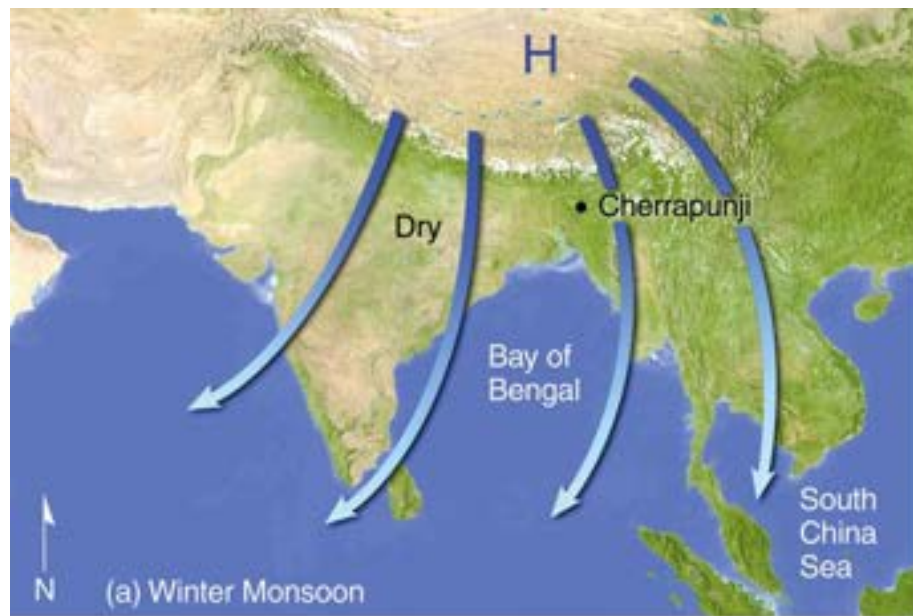
- is a seasonal change in the wind direction.
- Indian Summer Monsoon (ISM) occurs between June and September every year (Stable phenomenon).



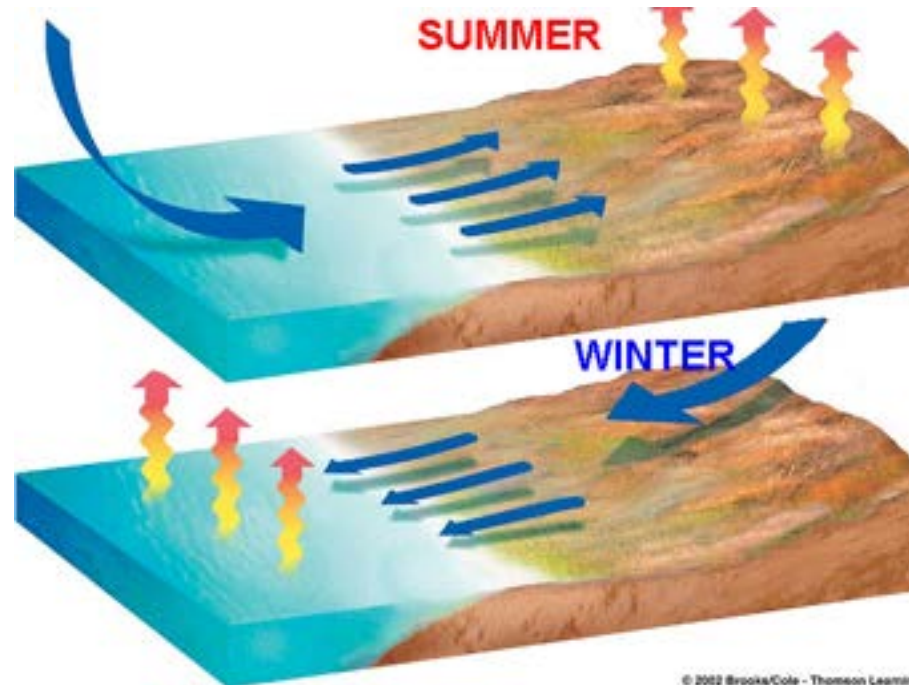
Why it is important?

- Because most of the population live in this region. Many of these societies rely on rain-fed agriculture for the food, and then it also controls the economy of the country.

- Water Resources
- Power generation

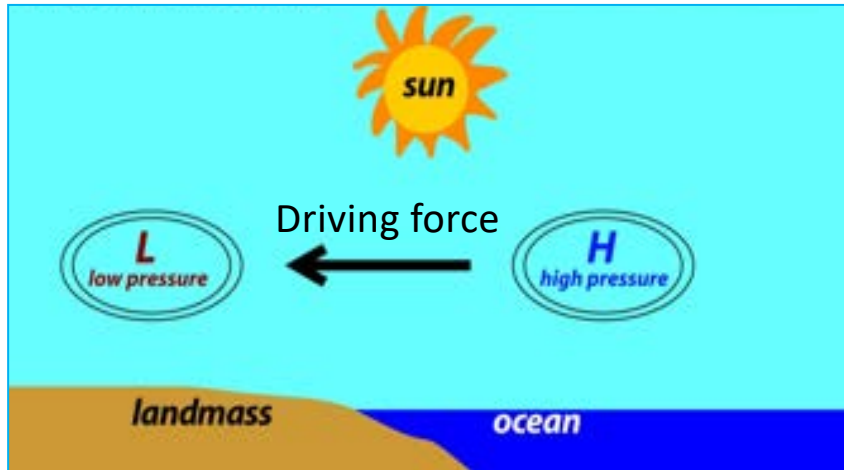


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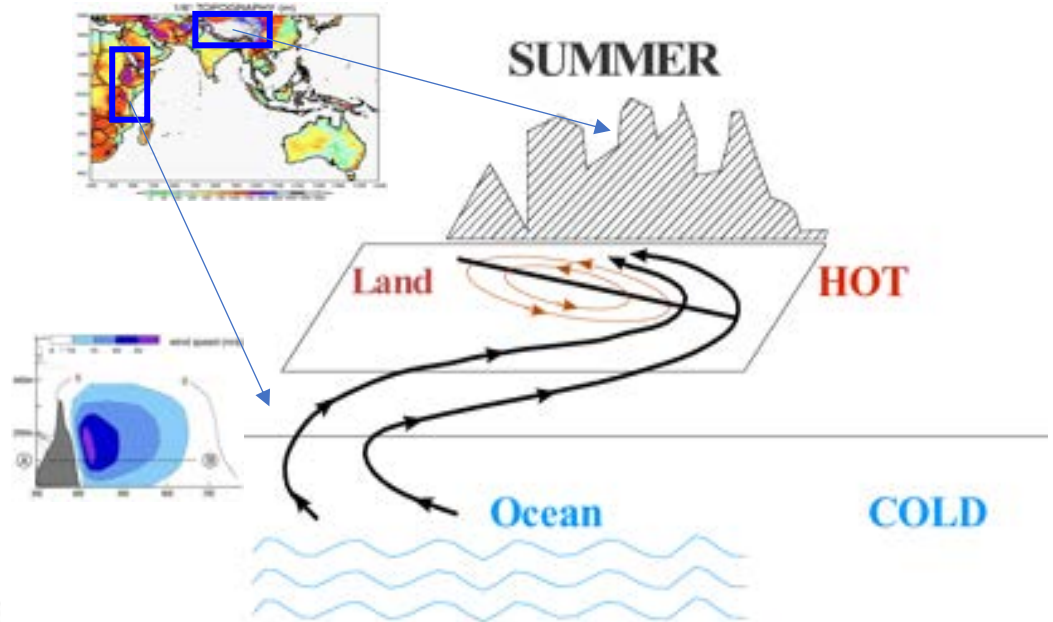
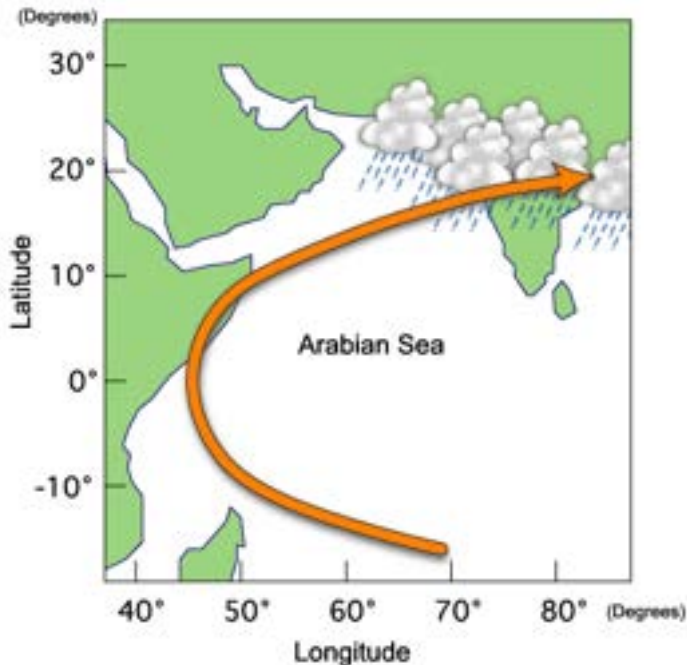


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How monsoon system forms ?

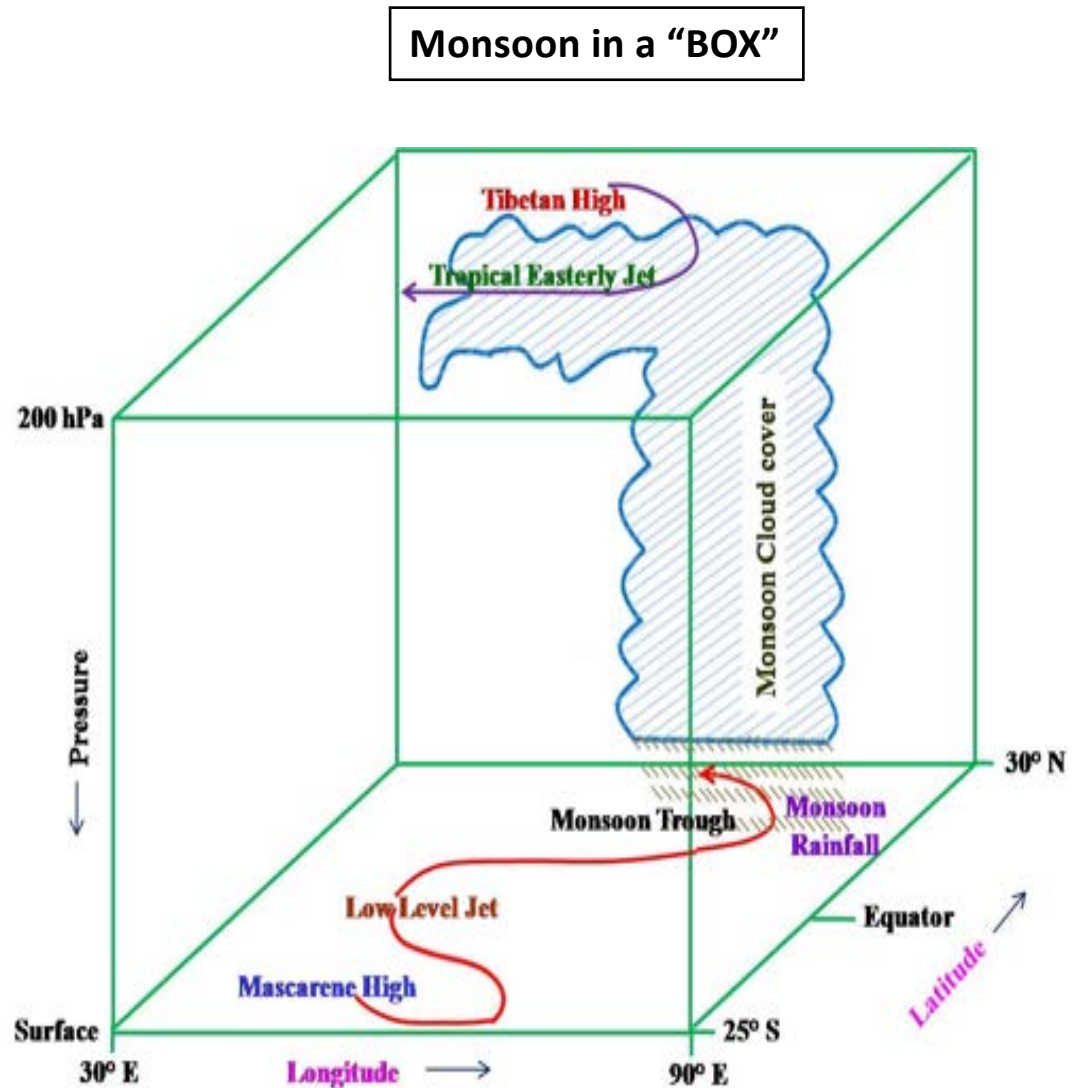


- ❖ Differential Heating
- ❖ Heat capacities of water and land is different.
- ❖ Difference between the specific heat of water ($4218\text{Jkg}^{-1}\text{K}^{-1}$) and land ($1300\text{Jkg}^{-1}\text{K}^{-1}$) is 3.3 factors.
- ❖ This heating set ups the pressure gradients.



Monsoonal Elements:

1. Monsoon Trough
2. Mascarene High
3. Low level jet stream (Somali jet)
4. Tibetan anticyclone
5. Tropical Easterly Jet
6. Monsoon Clouds
7. Monsoon Rainfall



(Inspired by “Krishnamurti and Bhalme, 1976)

Definition of the Indian Summer Monsoon

- ISM is a large scale land-sea breeze system caused by the thermal gradient between land (Indian sub-content) and sea (Indian Ocean).
- ISM is a seasonal manifestation of northward migration of the [ITCZ](#).
- It occurs between June and September every year.

Classical Criteria for the monsoon (Ramage 1971):

- ✓ Prevailing wind shifts 120° between January and July
- ✓ Average frequency of prevailing wind direction $> 40\%$
- ✓ Speed of mean wind should exceed 3 m s^{-1}
- ✓ Pressure patterns satisfy a steadiness criterion

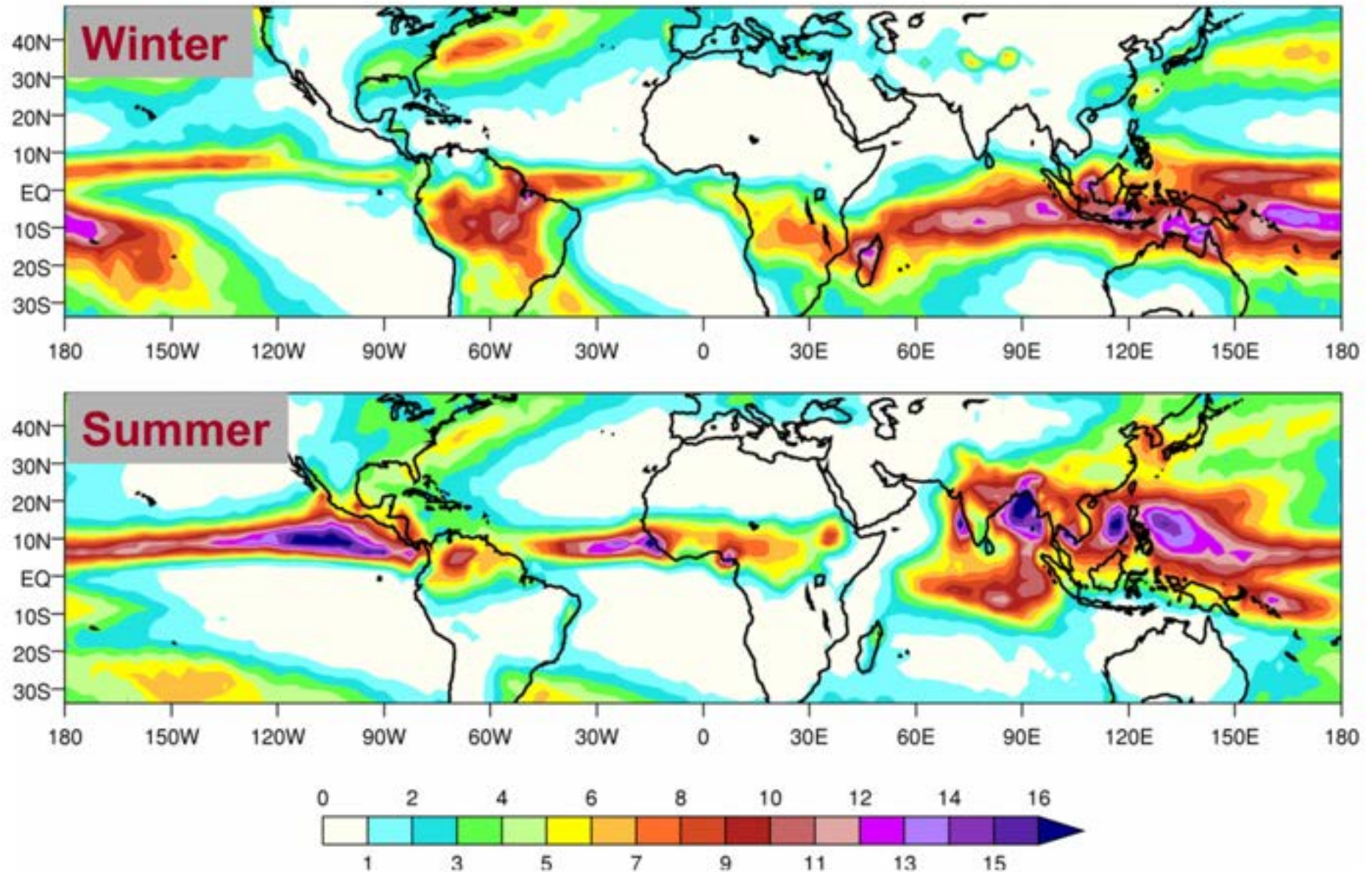
According to this criteria, the monsoon region is as



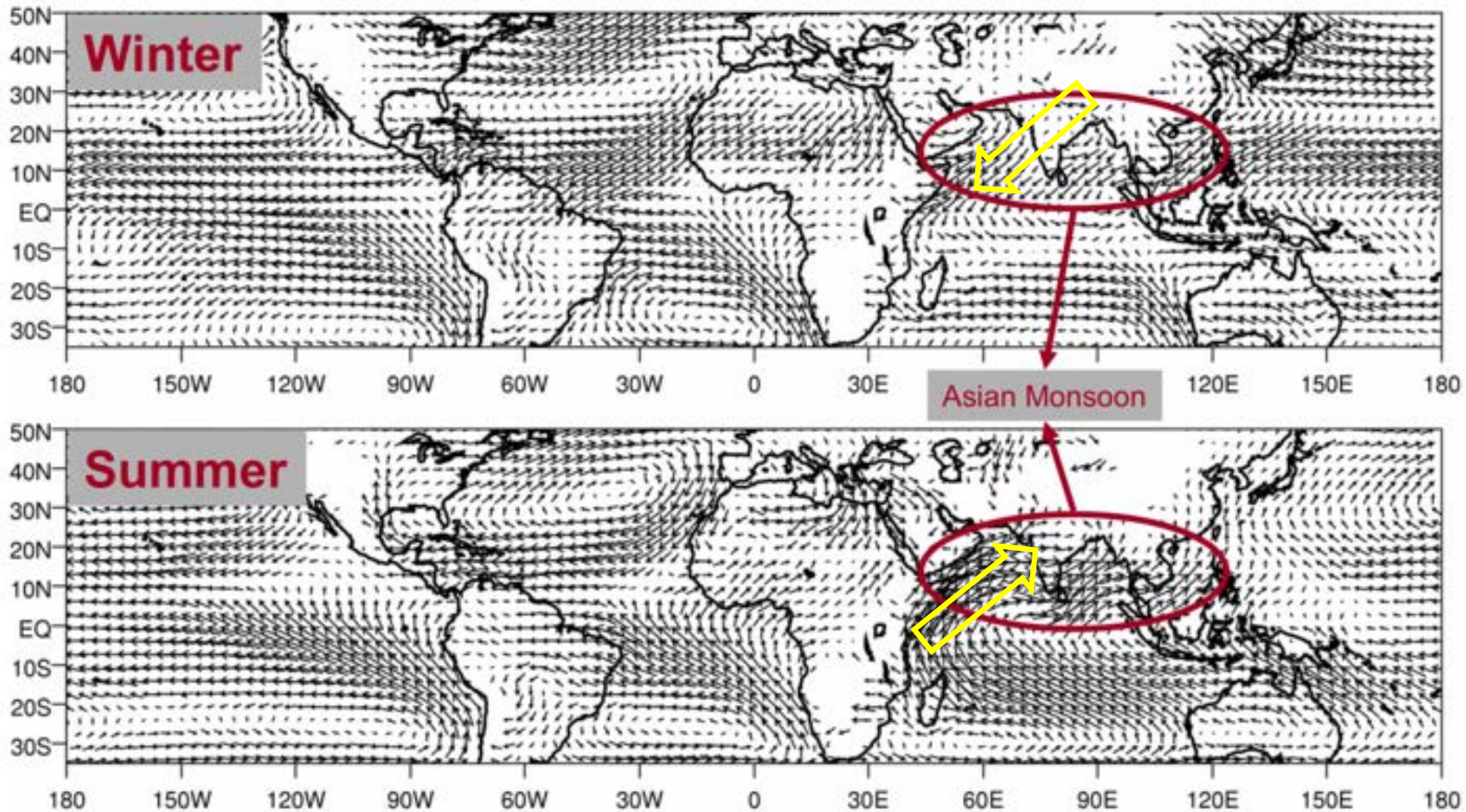
Typical Seasonal manifestation of ITCZ

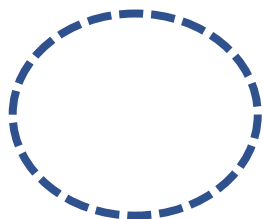


Seasonal manifestation of ITCZ



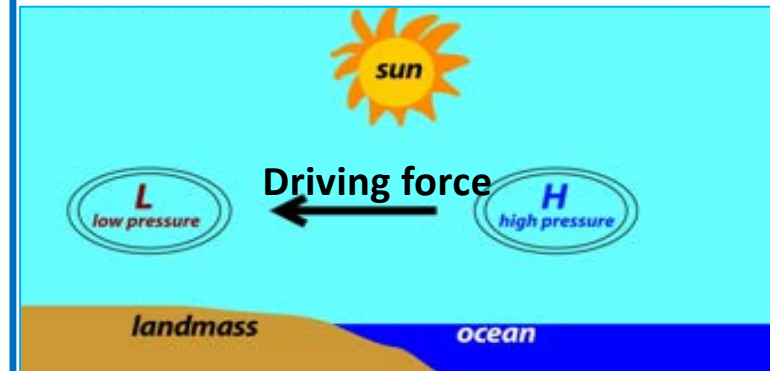
Surface wind patterns





Potential drivers of the monsoon:

- ❖ **Differential heating** between land and ocean, resulting pressure gradient (Halley 1686).
- ❖ Difference between the specific heat of water ($4218\text{Jkg}^{-1}\text{K}^{-1}$) and land ($1300\text{Jkg}^{-1}\text{K}^{-1}$) is 3.3 factors.
- ❖ This heating set ups the pressure gradients.



- ❖ Due to seasonal oscillation of Solar heating, **landward migration of ITCZ**

❖ **Earth's rotation**

Coriolis force drives monsoon circulation in curved paths



❖ **Moisture processes in the atmosphere**

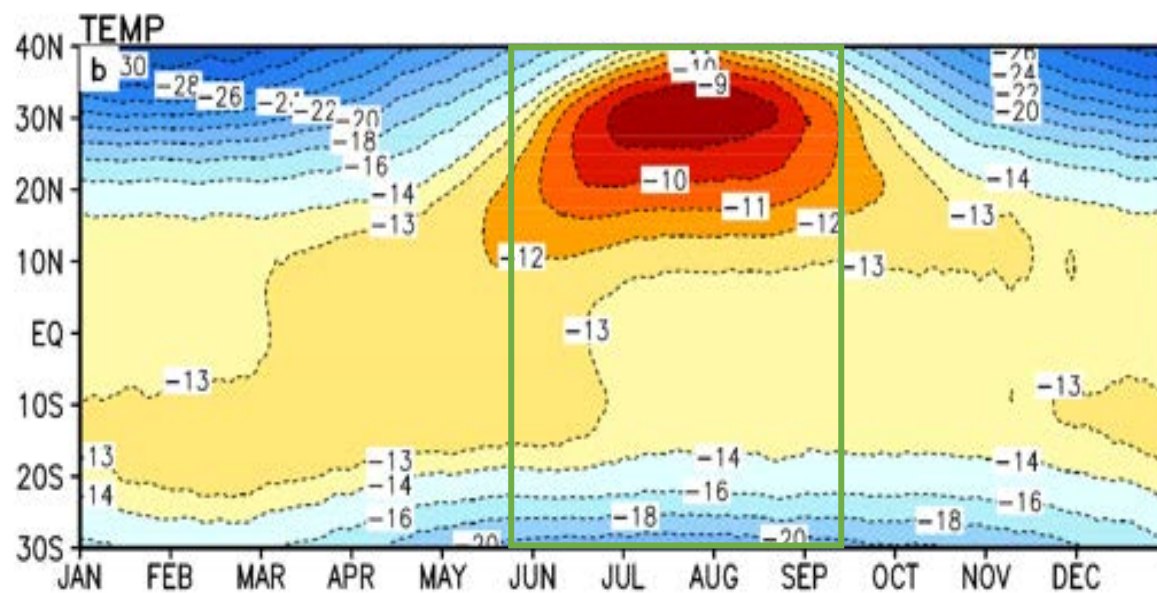
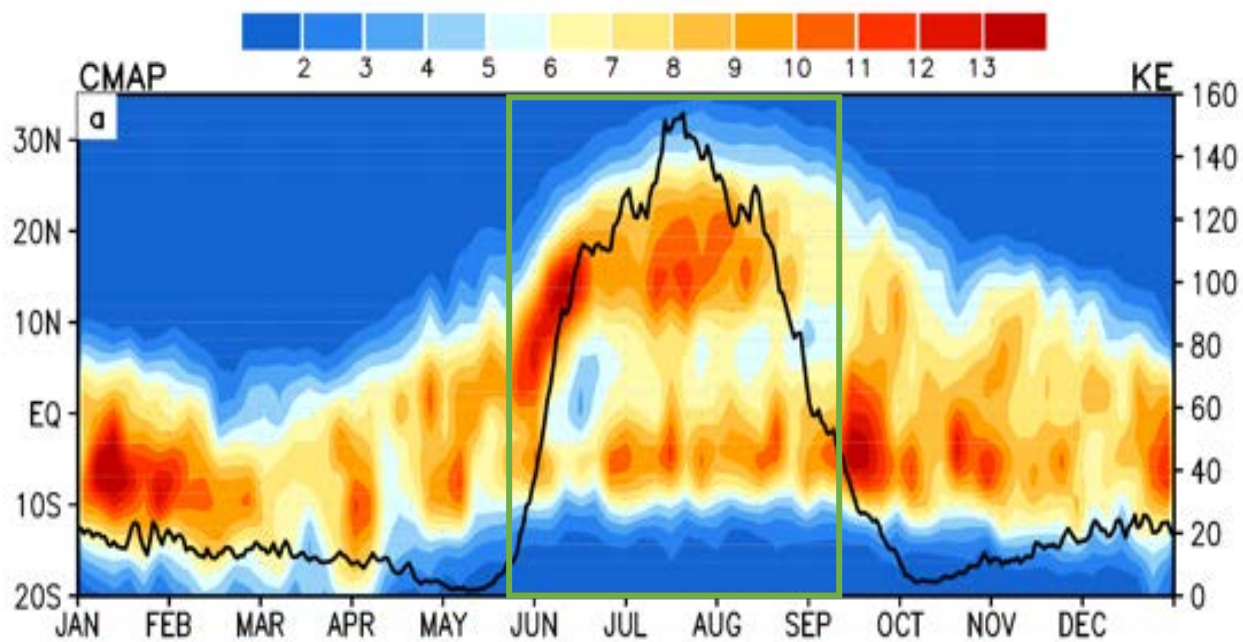
As moist warm air rises over summer time heated land surfaces (thermal instability), the moisture eventually condenses, releasing energy in the form of latent heat of condensation. This energy maintaining summer land-ocean pressure gradients adding to monsoon over a season

Thermal Contrast concept- Cannot Explain:

- Vertical structure of circulation: Low level conv. , upper level div.
- Also, after the initial rainfall, the land temperature is actually cooler than that of the ocean

ROLE OF MOIST PROCESSES

- Differential heating is not enough to explain the strength and extent of monsoon circulations.
- Moisture acts as “stored energy” through latent heat release.
- Evaporation occurs over the oceans, and then moisture is transported over the land, where it is released through condensation.
- Latent heating results in a more intense monsoon flow, and also a vertically deeper monsoon flow.
 - A moist monsoon has a depth on the order of the troposphere.
 - A dry monsoon is much shallower, extending only to the mid-troposphere.



Monsoonal Elements

Semi-permanent features of the monsoon

1. Heat Low and Monsoon Trough (**Lower atmosphere- Surface to 850 hPa**)
2. Cross-Equatorial Flow: Low level (Somali) jet stream (**Lower atmosphere**)
3. Mascarene High (**Lower atmosphere**)
4. Tibetan anticyclone (**Upper atmosphere**)
5. Tropical Easterly Jet (**Upper atmosphere**)
6. Monsoon Clouds
7. Monsoon Rainfall

Monsoonal Elements: 1. Heat Low and Monsoon Trough

- The intense heating over northwest India in summer results into the development of a low pressure area over the region.
- It is an extension of intertropical convergence zone (ITCZ).
- It characterized by the ascent of the wind only up to 850 hPa pressure level (roughly 1.5 km), above that air subsides over the heat low - Shallow Cell.
- Heat low is response of dry atmosphere driven by the temperature at the surface (dry convection).
- Therefore no rainfall.

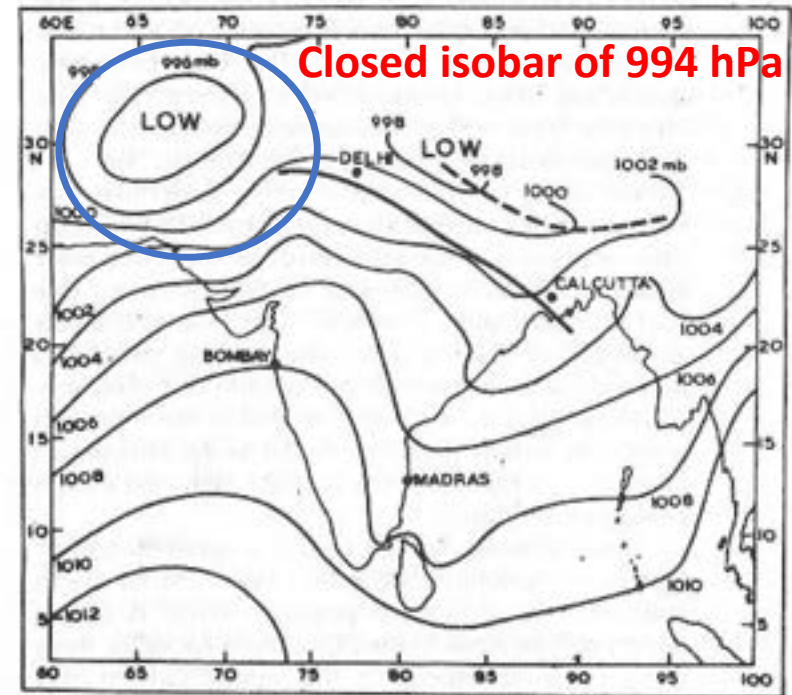


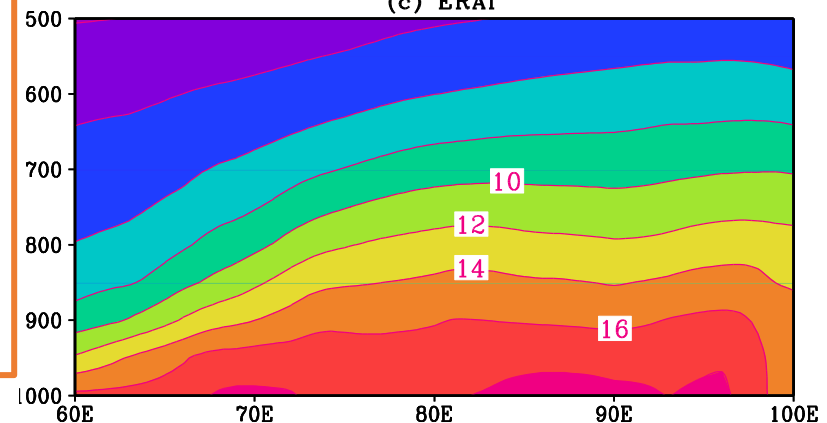
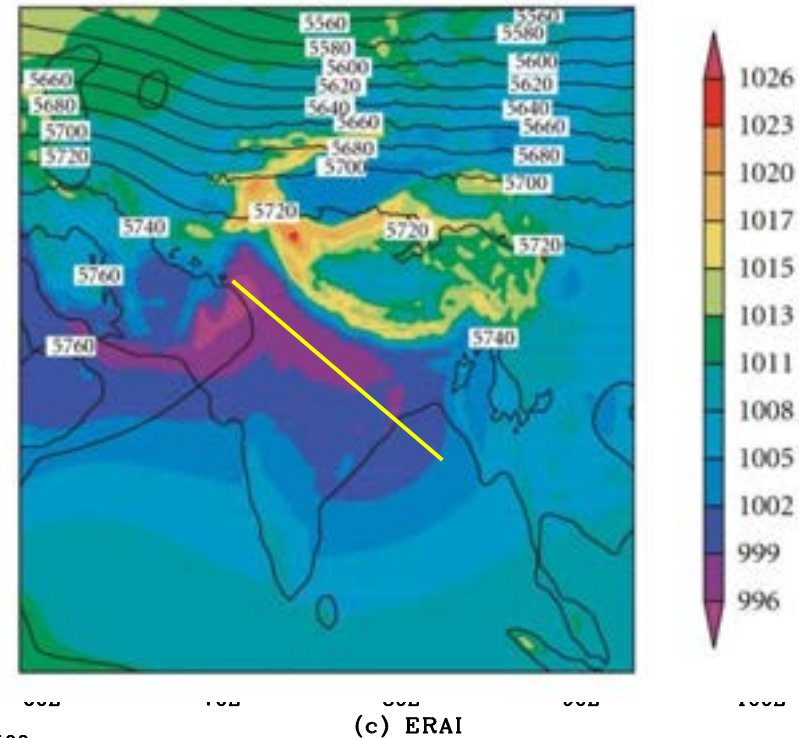
Figure 3.3. Sea-level weather chart for a typical 'break' monsoon. Normal position of the axis of monsoon trough during an active monsoon phase is shown by a thick continuous line. Dotted line indicates its position during a 'break'.

Mean Pressure in summer

Monsoonal Elements: 1. Heat Low and Monsoon Trough

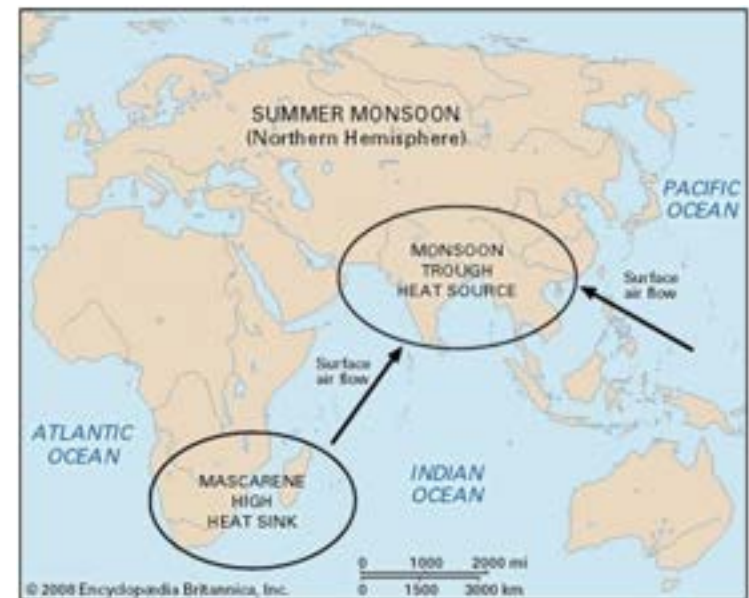
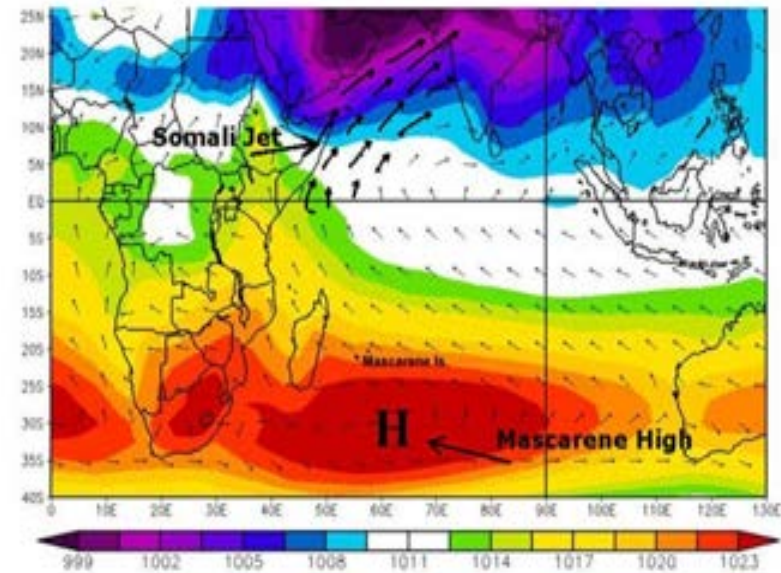
The most important aspect of Indian monsoon (Dynamical Heat low).

- An east-west oriented low pressure, stretching across the Indo-Gangetic plains towards Bay of Bengal.
- Often associated with heavy rainfall
- North-South displacement of trough determines the rainfall
 - Active monsoon- Southward shift
 - Break monsoon- Northward shift
- The crux of the monsoon performance is depends on the dynamics of the monsoon trough
- Vertically Deeper



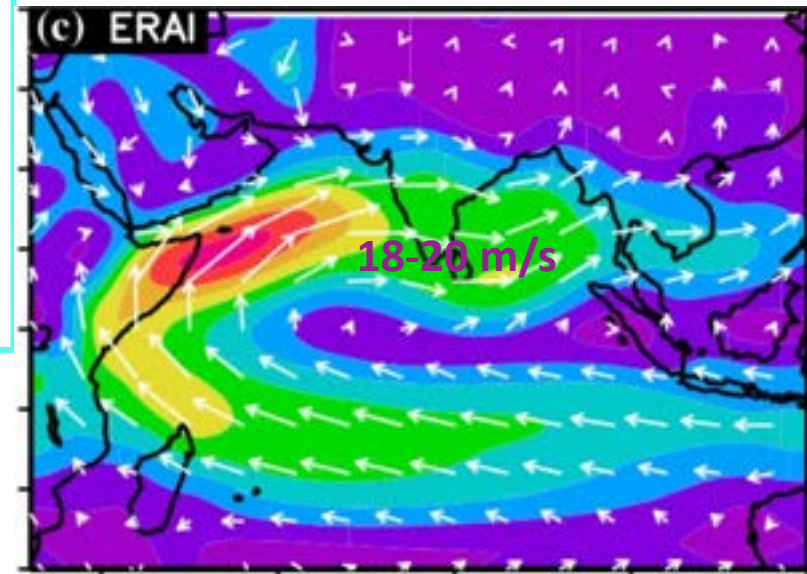
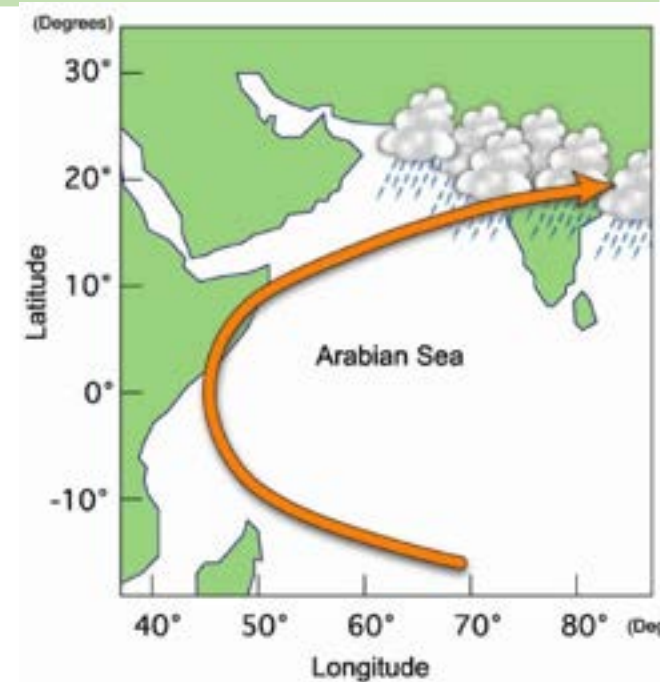
Monsoonal Elements: 2. Mascarene High

- Mascarene high is characterized by high pressure center over south of the equator.
- This high-pressure region is located between 25°S - 35°S and 40°E - 90°E near the Mascarene Islands in the southern Indian Ocean.
- It is the source of monsoon winds (heat sink)
- If there is a delay in the formation of Mascarene High, there is also the possibility of a delay in the onset of monsoon in India.



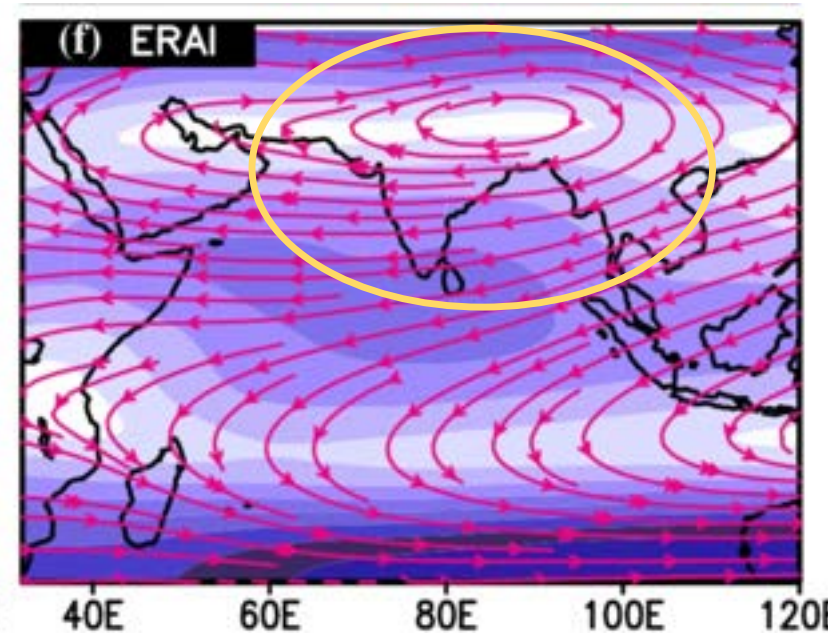
Monsoonal Elements: 3. Cross-Equatorial Flow/ Low level jet stream

- Large outflow of air takes place from this area and it crosses the equator as cross-equatorial Somali jet and becomes south-westerly wind.
- Low Level Jet is also known as Findlater jet or Somali jet, it is a narrow channel of wind with maximum winds near 1.5 km height level off the coast of Somali.
- The Somali jet is one of the most prominent wind systems of the lower troposphere during the Indian summer monsoon.
- Max speeds (18-20 m/s) at the core of LLJ



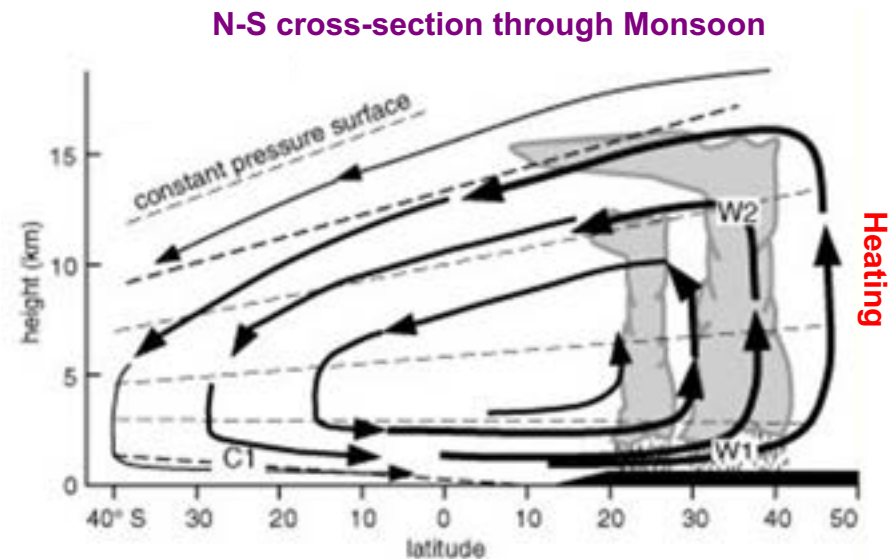
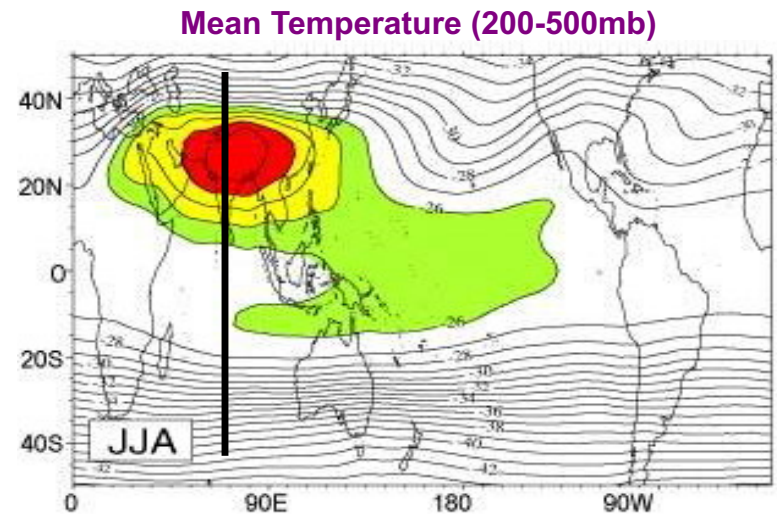
Monsoonal Elements: 4. Tibetan anticyclone/Tibetan high

- Most remarkable circulation system in the upper troposphere (200 hPa) during summer.
- There is a large anticyclone (a warm high pressure zone) having its largest amplitude near 200 hPa during the summer monsoon over the Tibetan plateau (height of ~ 4 km from the mean sea level).
- This upper level anticyclone system that forms above the monsoon trough located over north India.
- Formed due to sensible heating, a strong thermal wind rises and diverges at 200 hPa leads to anticyclonic circulation.
- The Tibetan plateau is considered to be one of the key factors in the development of monsoon circulation in the region.



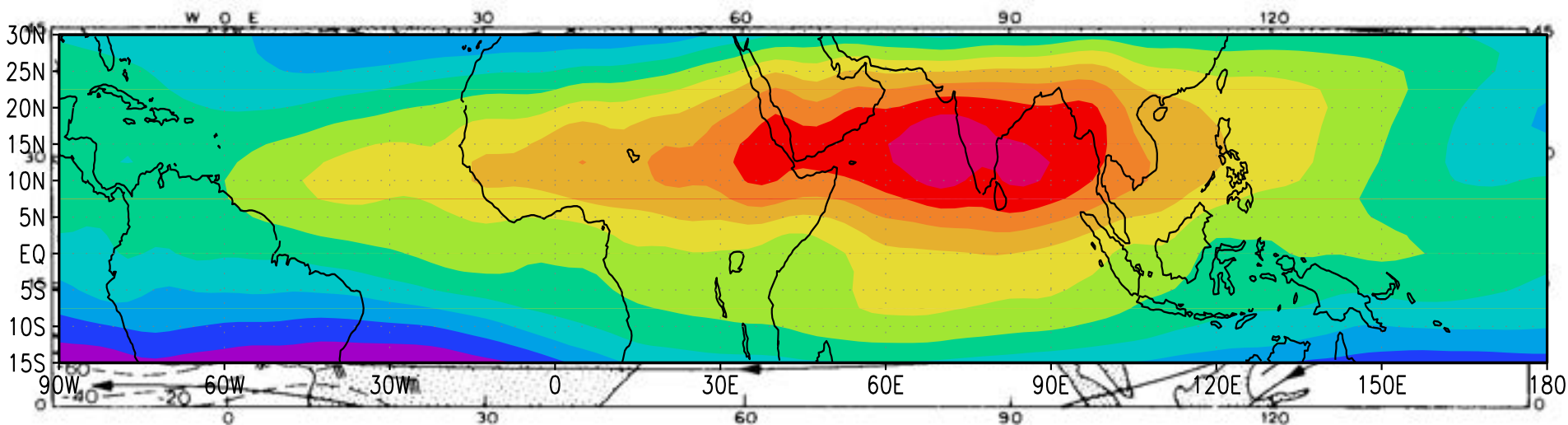
Monsoonal Elements: 4. Tibetan anticyclone/Tibetan high

- **Solar heating** of the Himalayas is quickly converted to **mid-level atmospheric heating** via sensible heat fluxes
- Mid-level heating increases the thickness between pressure surfaces: sets-up a strong pressure gradient at upper-levels and strong offshore flow
- Lowers surface pressure over land and induces onshore low-level flow that gains moisture from the ocean via surface fluxes
- Moisture convergence and forced ascent over land produces deep convection and **latent heat release**
- Both heat sources continue to drive the monsoon circulation

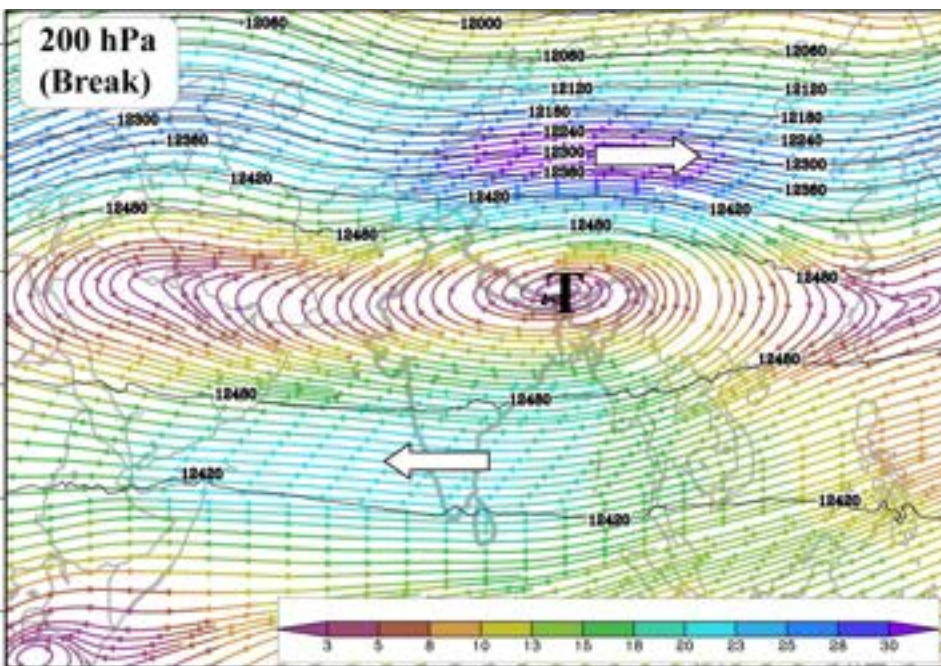


Monsoonal Elements: 5. Tropical Easterly Jet

- ✓ **The Tropical Easterly Jet** is an upper level easterly wind that forms in summer. This strong flow of air that develops in the upper atmosphere during the Asian monsoon is centered on 15°N , $50\text{--}80^{\circ}\text{E}$ and extends from South-East Asia to Africa.
- ✓ The strongest development of the jet is at about 15 km above the Earth's surface with wind speeds of up to 40 m/s over the Indian Ocean.
- ✓ Jet formed due to intense **heating over Tibet Plateau**.
- ✓ Formation: Thermal gradient is reversed, resultant wind is the tropical easterly jet.

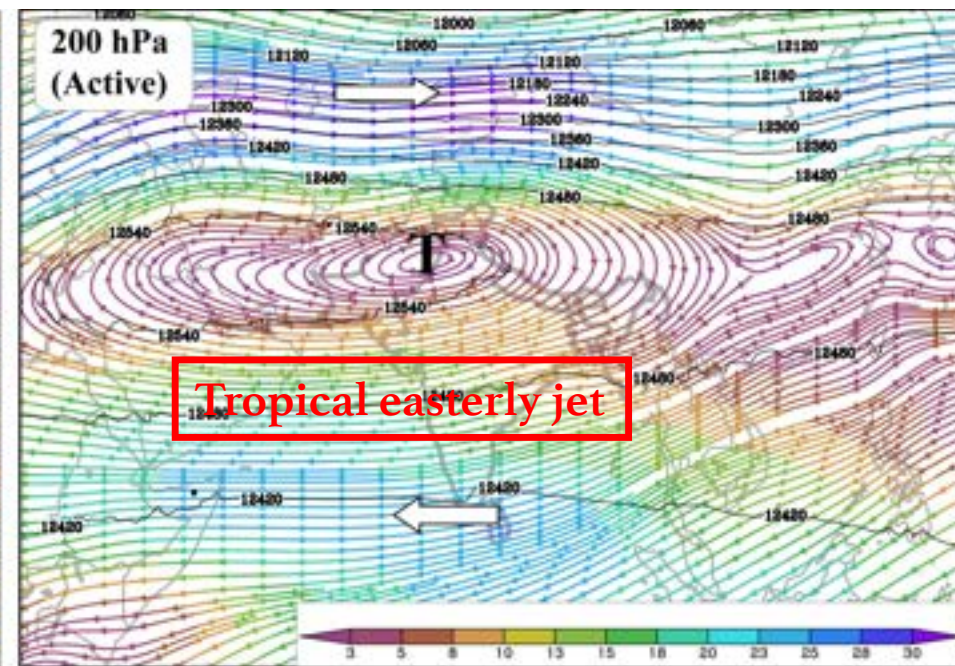


Tropical Easterly Jet during monsoon



Winds and geopotential height at 200 hPa

Weaker TEJ

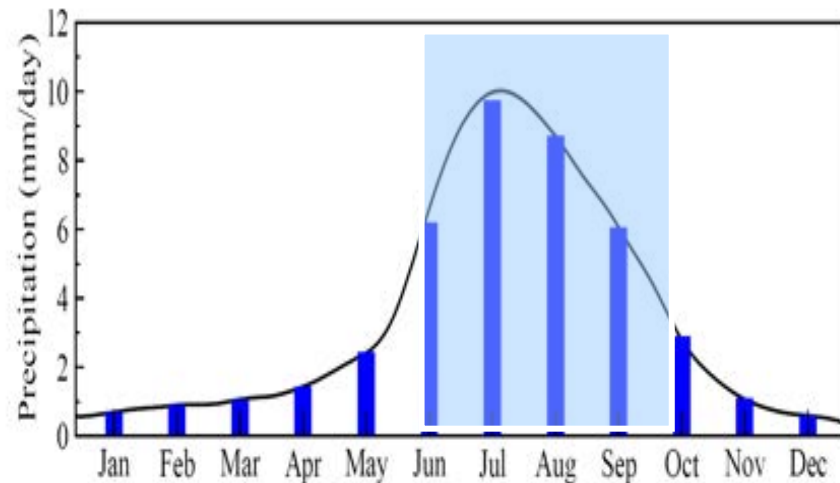


T = Tibetan anticyclone

Stronger TEJ

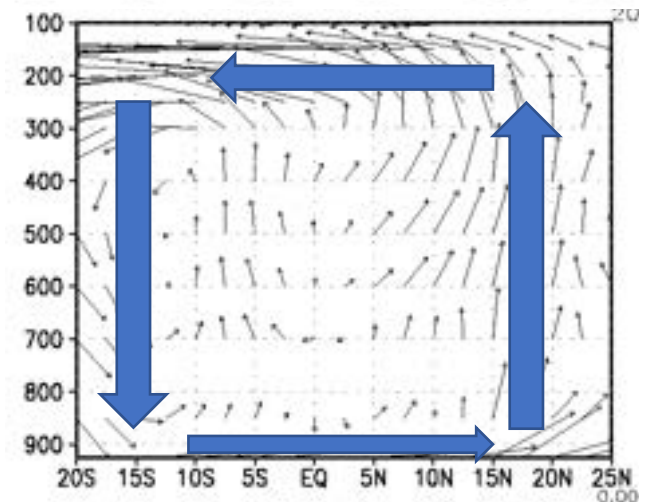
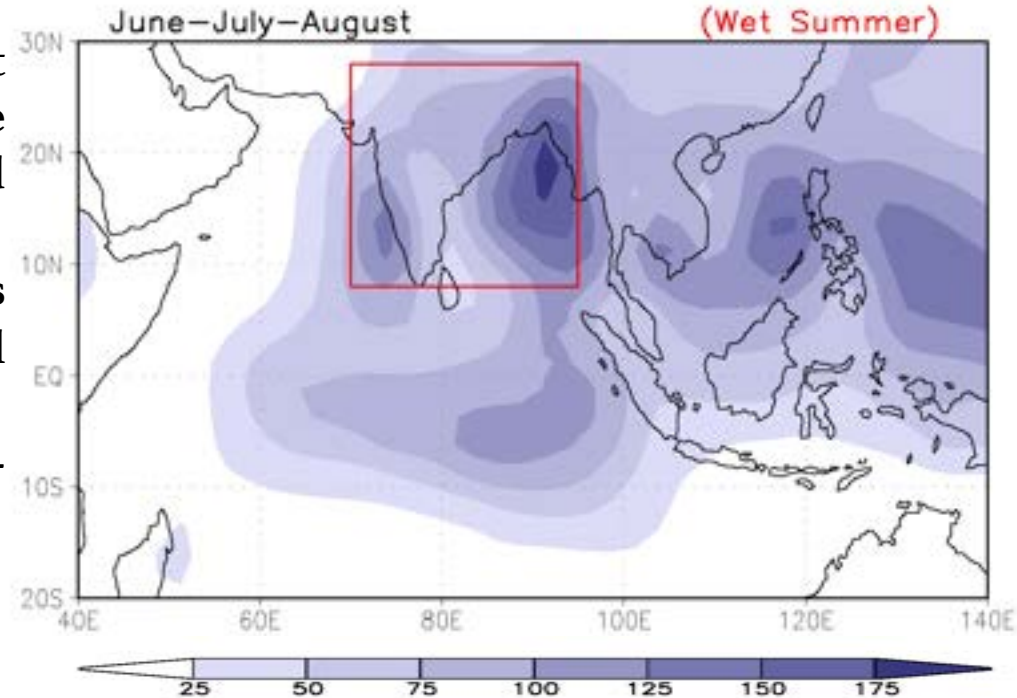
Monsoonal Elements: Rainfall

- High rainfall occurs along the West Coast of Indian (associated with the orography parallel to the coast) and over the north-eastern regions.
- Rainfall occurs over convective zones (Arabian Sea, Head BoB and Central Indian region).
- Rain shadow region over the South-eastern Peninsula



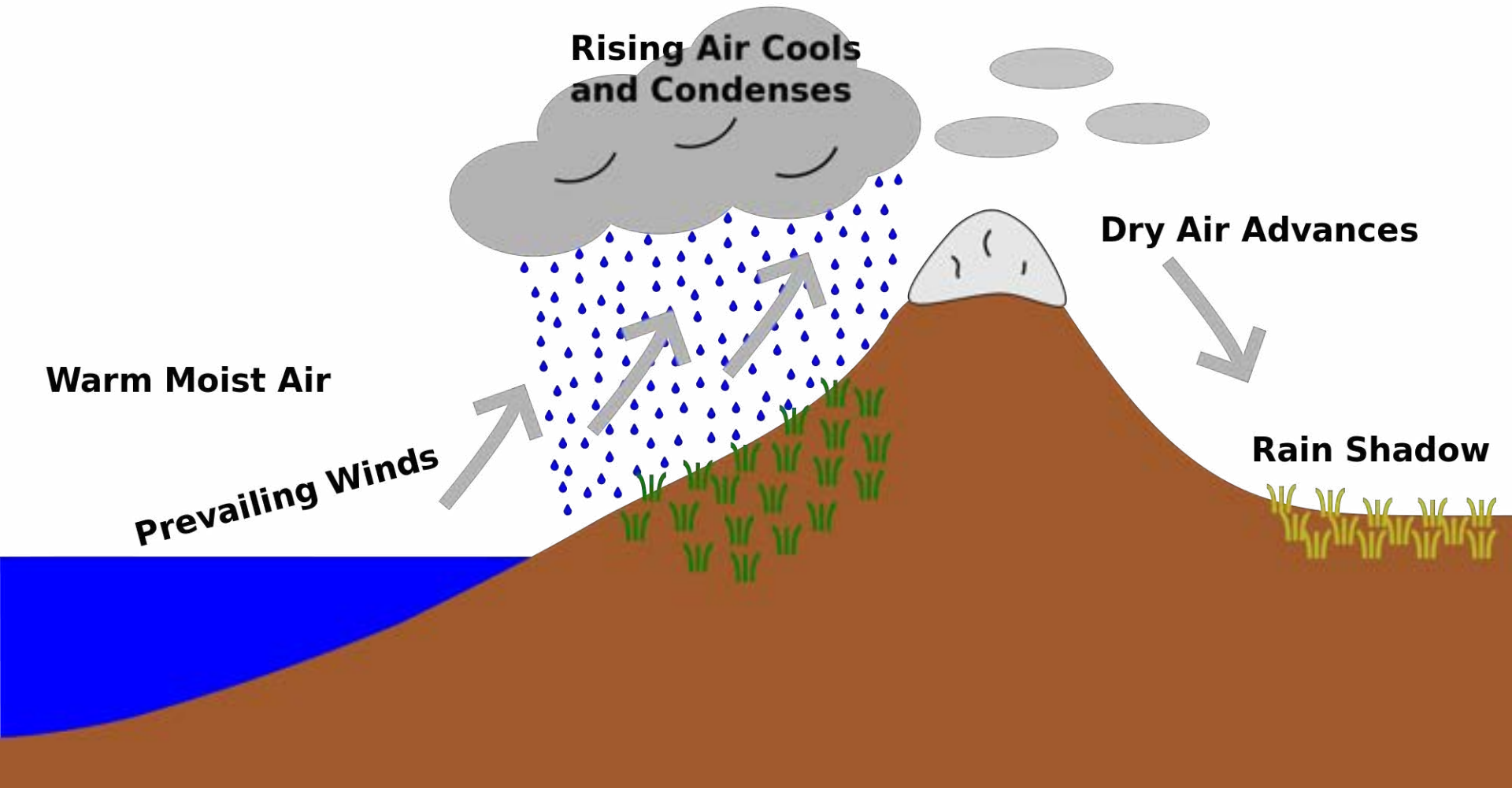
Annual Cycle of the Rainfall

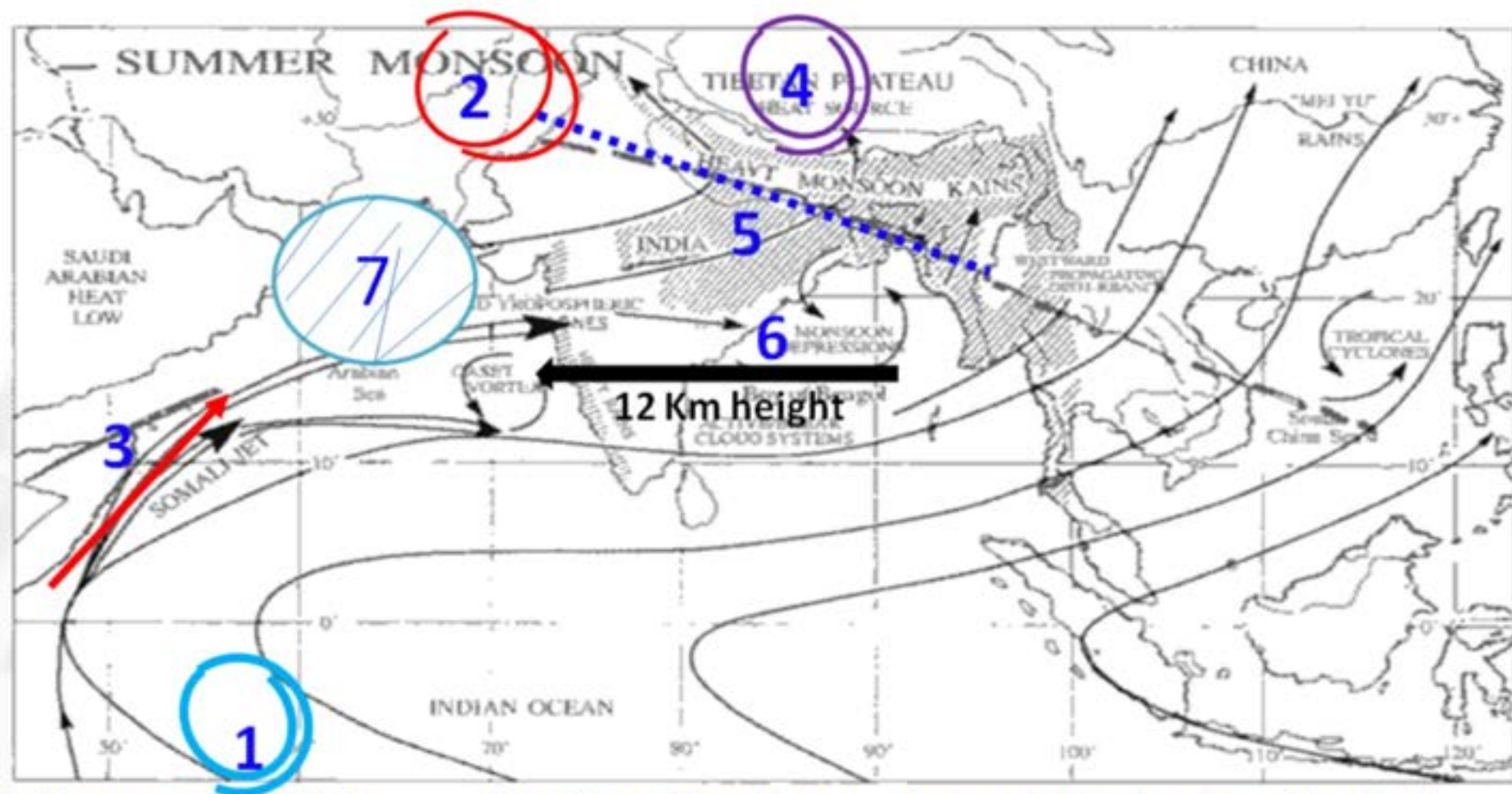
Long term mean summer precipitation



Monsoon Hadley cell

Wind- & Lee- ward





1) Mascarene high

2) Heat Low

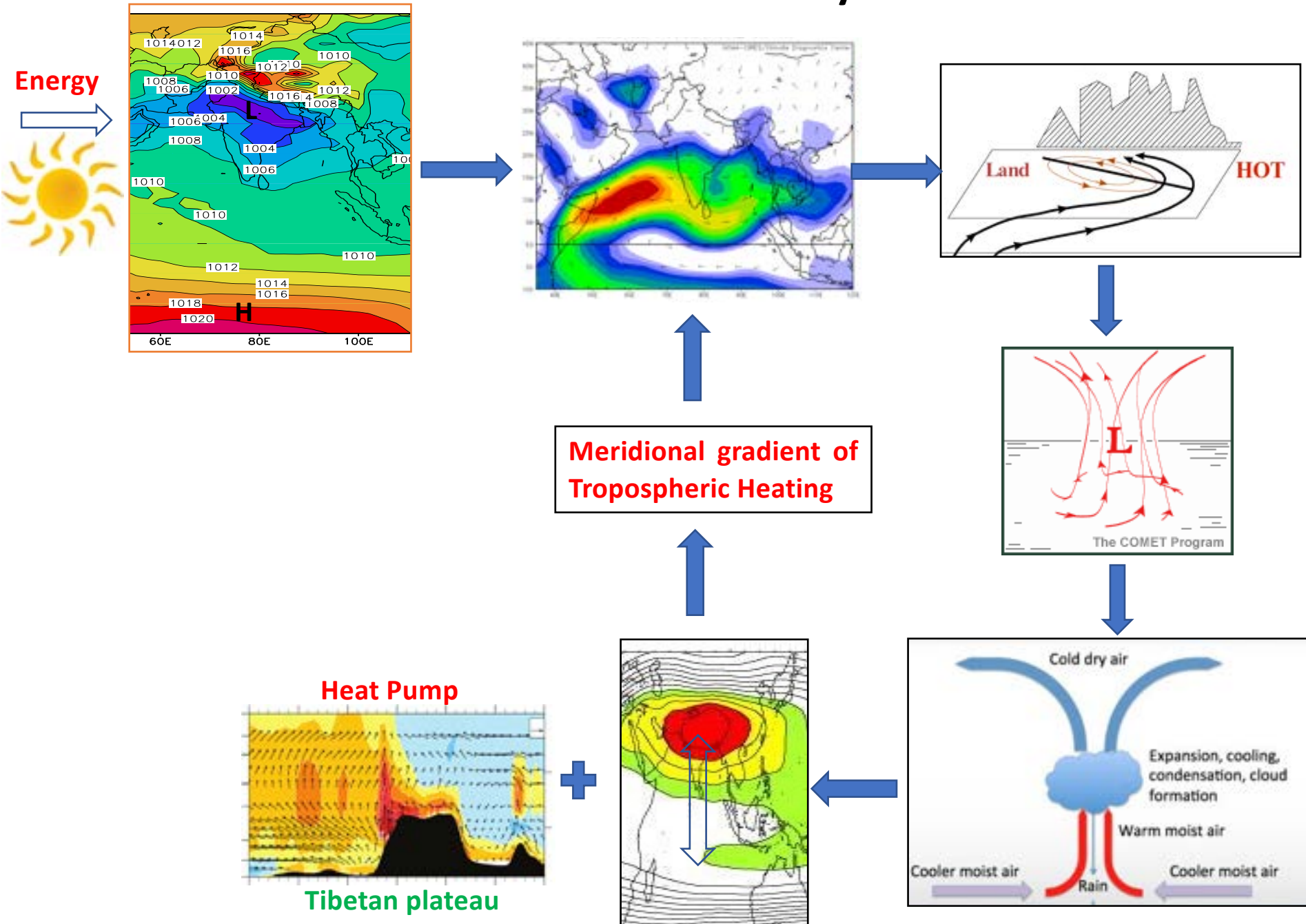
3) Somali low level Jet (~ 1 km)

4) Tibetan high (~ 5 km)

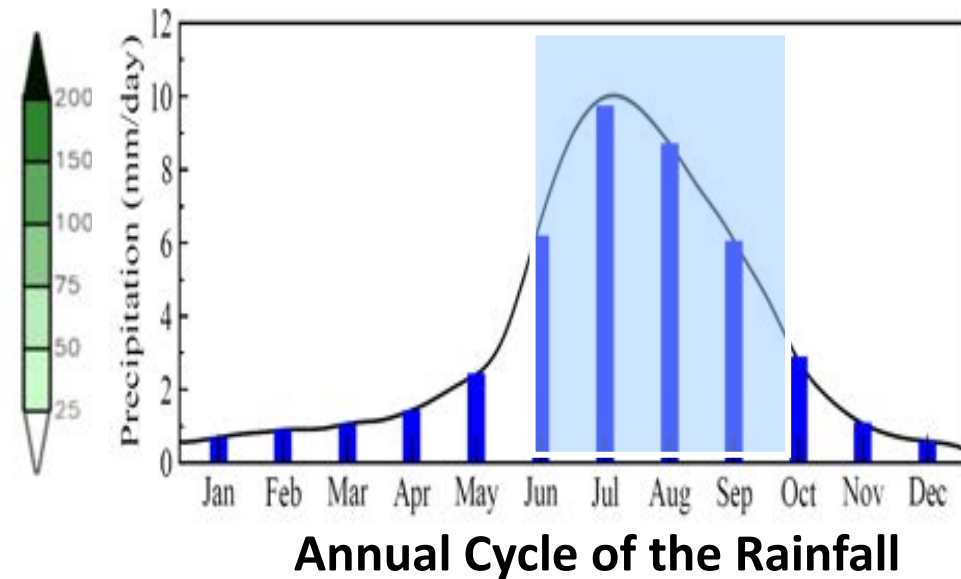
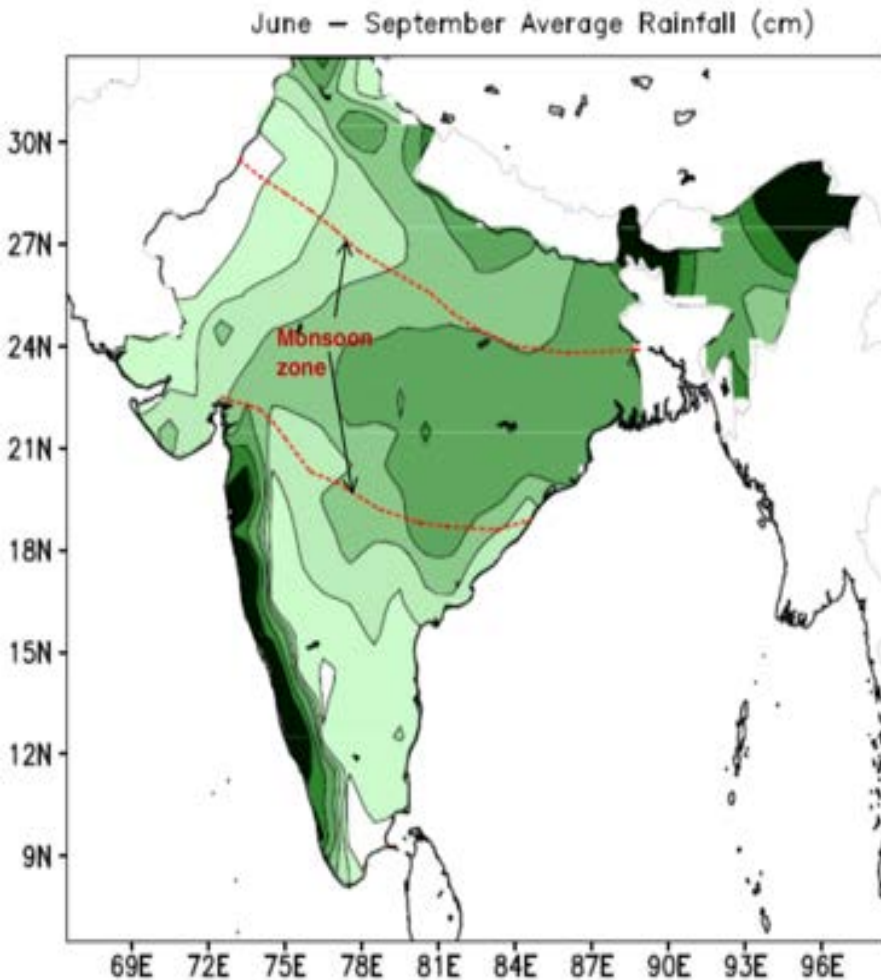
5) Monsoon Trough

6) Tropical easterly jet (~ 12 km)

Monsoon Theory



Synoptic Weather Systems of the Summer Monsoon

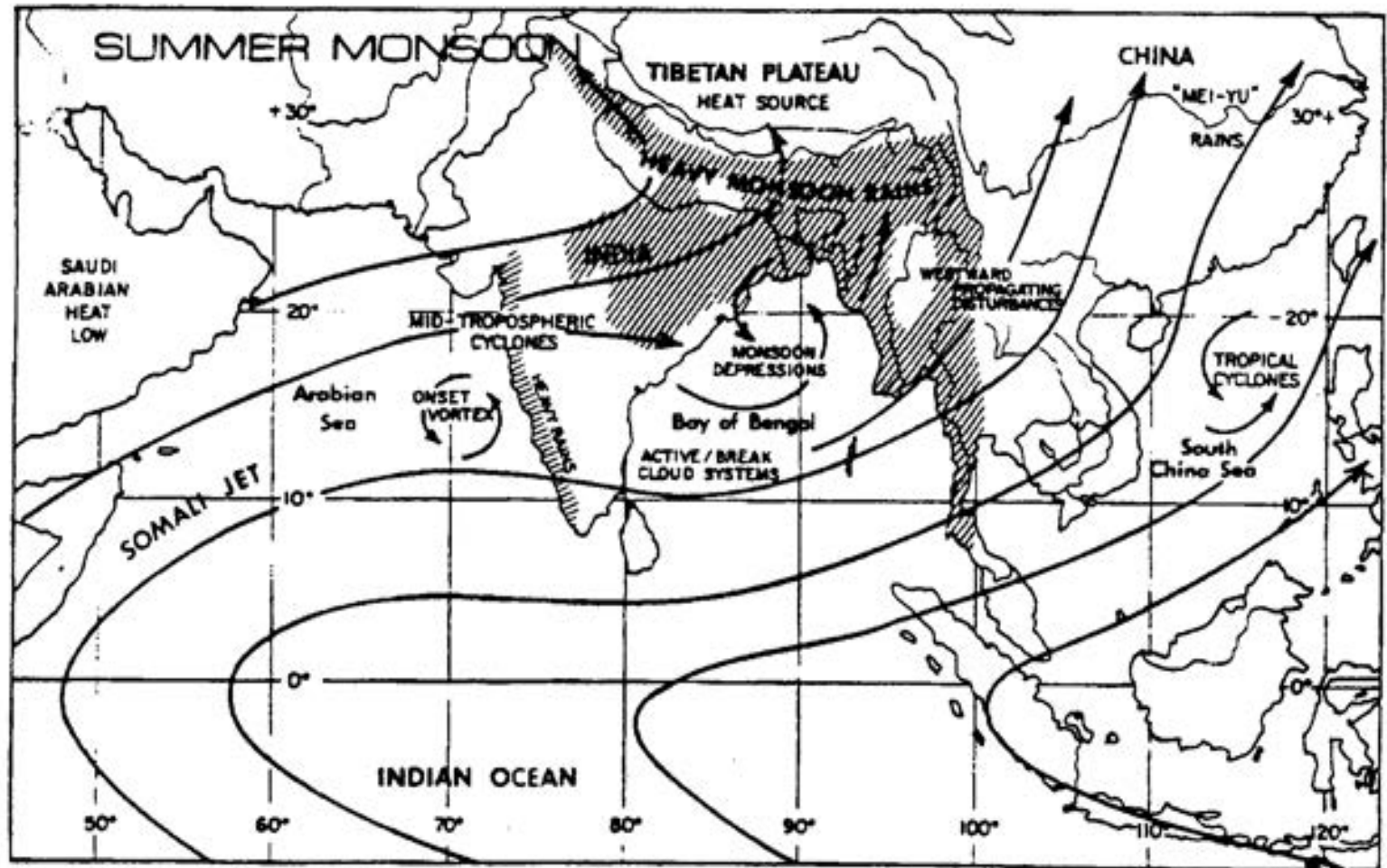


Spatial Distribution of Long term mean Summer Rainfall

Synoptic Weather Systems of the Summer Monsoon

- ❖ Monsoon Trough
- ❖ Monsoon Lows/Depressions (Low Pressure Systems)
- ❖ Offshore Troughs
- ❖ Mid-tropospheric Cyclones
- ❖ Western Disturbances
- ❖ Orography*

Synoptic Weather Systems of the Summer Monsoon

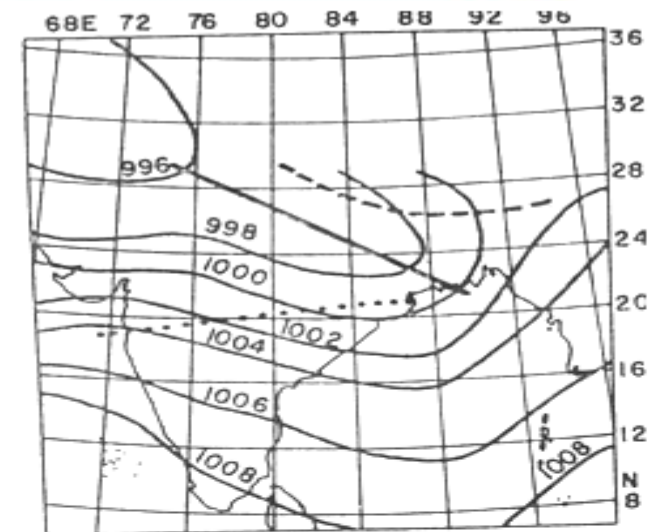
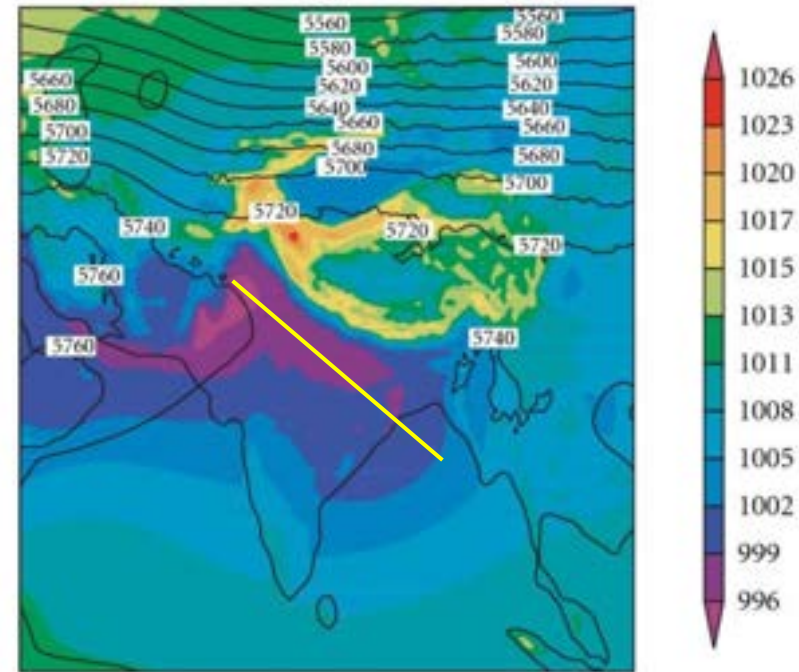


Hatched region for
RR>1000 mm
Source : Rao, 81

Monsoon Trough

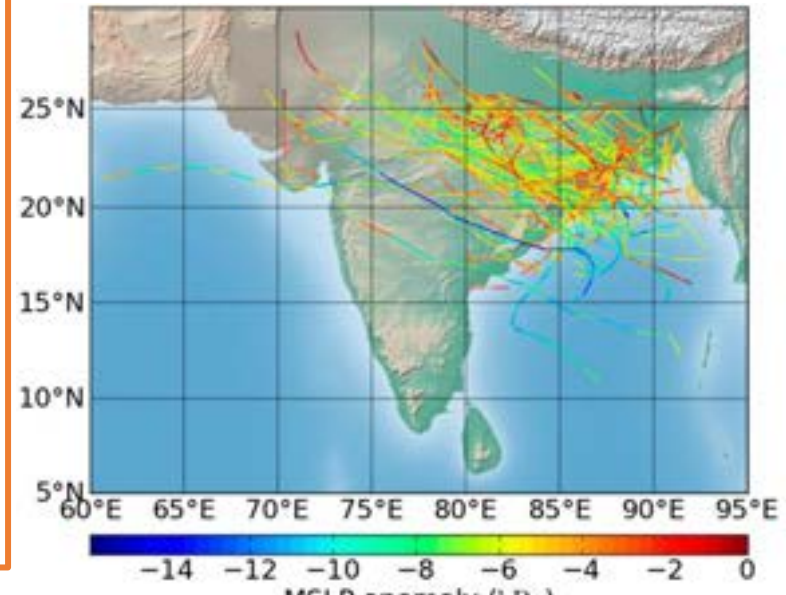
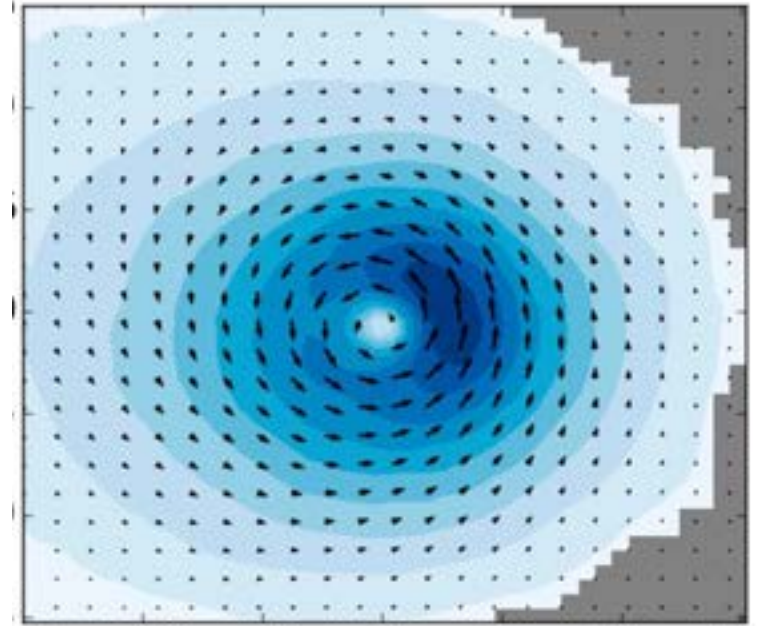
Most dominated feature of the monsoon.

- An east-west oriented low pressure, stretching across the Indo-Gangetic plains towards Bay of Bengal.
- Often associated with heavy rainfall over the Central Indian region
- North-South displacement of trough determines the rainfall location;
Active monsoon- Southward shift
Break monsoon- Northward shift
- Vertically Deeper
- Monsoon trough and its oscillation manifested in the midseason (July & August): active and break spells
- Life span: 10 to 30 days
- The intraseasonal pattern of the monsoon trough has a dominant 3-5 day synoptic scale oscillation.



Monsoon Lows/Depressions

- Monsoon Low/Depression is the disturbance which is associated with 2-3 closed isobars (at 2 hPa interval) and is associated windspeeds in the range of 18 to 33 knots (8.5 to 18 m/s).
- When the maximum surface wind speeds exceed 33 kts, then they become Cyclonic storms.
- Usually forms over seas but can also form over land regions (Land Depressions).
- Horizontal scale : about 1000 - 3000 km of diameter
- In the Monsoon season, most of the depressions form in the BoB compared to Arabian Sea.
- Movement of these low/depressions: Westward/Northwestward.
- Rainfall contribution: Core monsoon region/Eastern Indian region

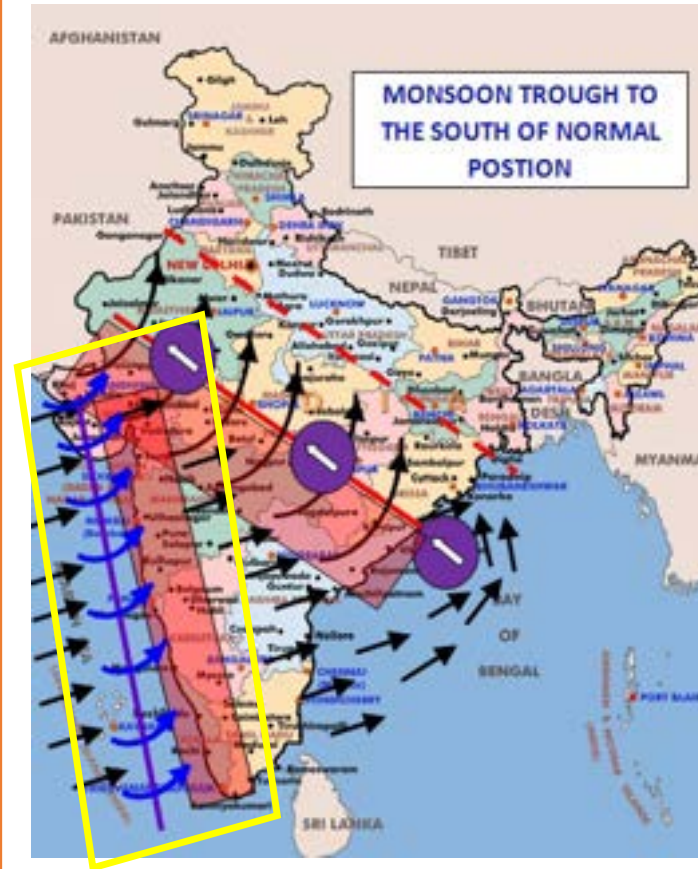


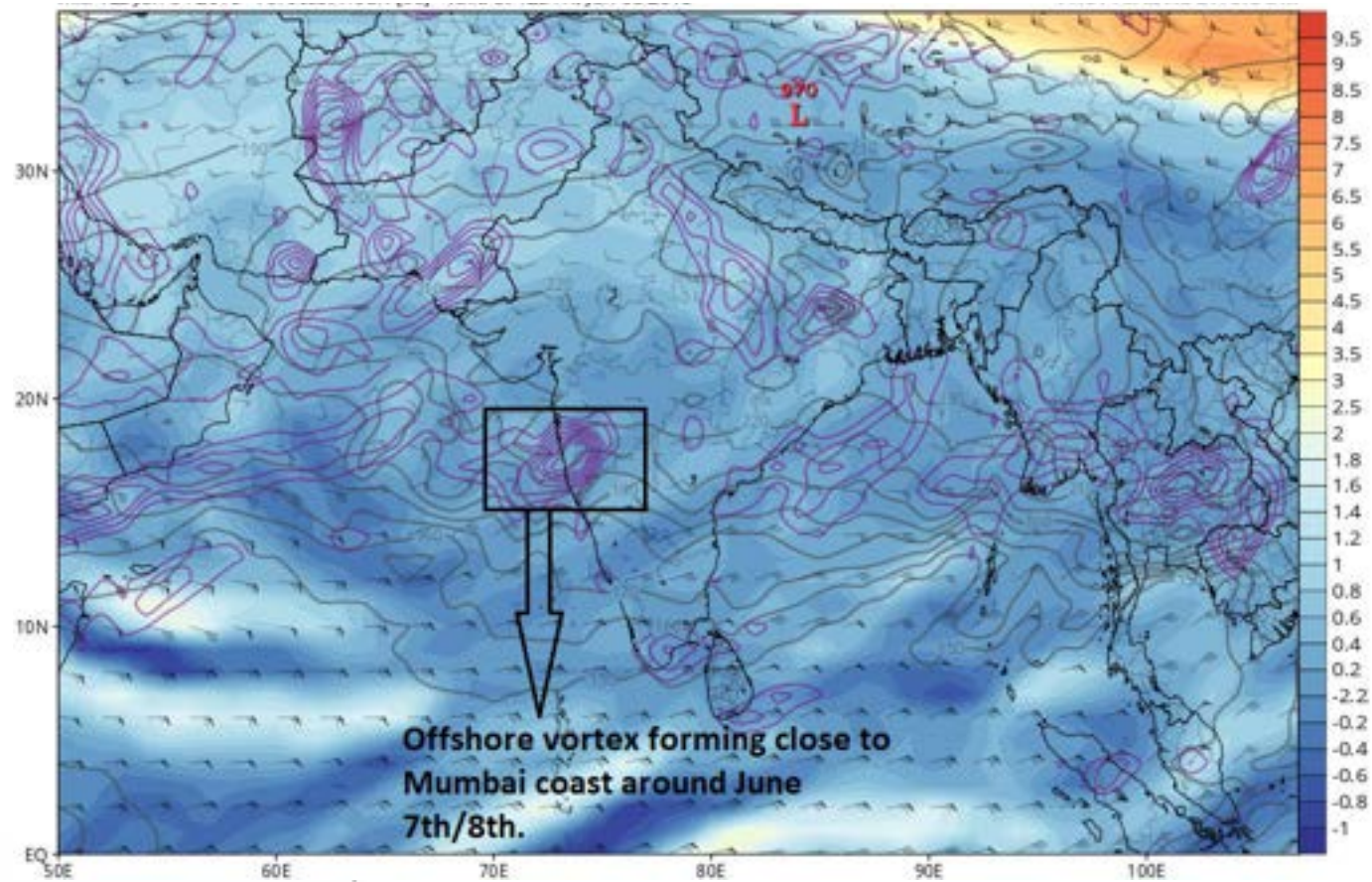
Frequency and Life Span of Monsoon Lows/Depressions

- Horizontal scale : about 1000 - 3000 km of diameter
- Location of initiation
 - ❖ 80 % in the Bay of Bengal
 - ❖ 10% in Arabian Sea
 - ❖ 10 % over land (Bangladesh)
- MSLP ~ as low as 990 hPa
- Life span : from 3 to 7 days.
3 (5) days for Land (Ocean) depressions
- Frequency : Twice a month over Bay of Bengal
- Movement : Westward or Northwestward at 2 or 3 m/s in direction of trough

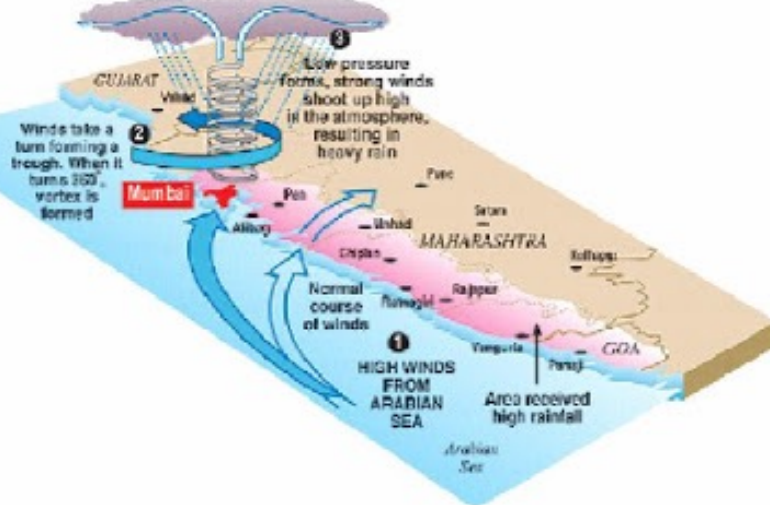
Offshore Troughs

- It is another important synoptic configuration is witnessed in the form of south to north oriented trough near the surface along the west coast of India.
- Offshore trough gives heaviest rainfall (10 cm per day) in the Western Ghats but reduces lee side of the western Ghats.
- Life span of the he offshore trough is about 3-7 days, with embedded mesoscale vortices of 1-2 days duration
- Horizontal scale of 200 km
- Vertical extent is about 3 km
- Preferred zones: Coastal Karnataka; Goa; Konkan and Gujarat
- Latent heat and orography (Western Ghats) causes enhanced the rainfall

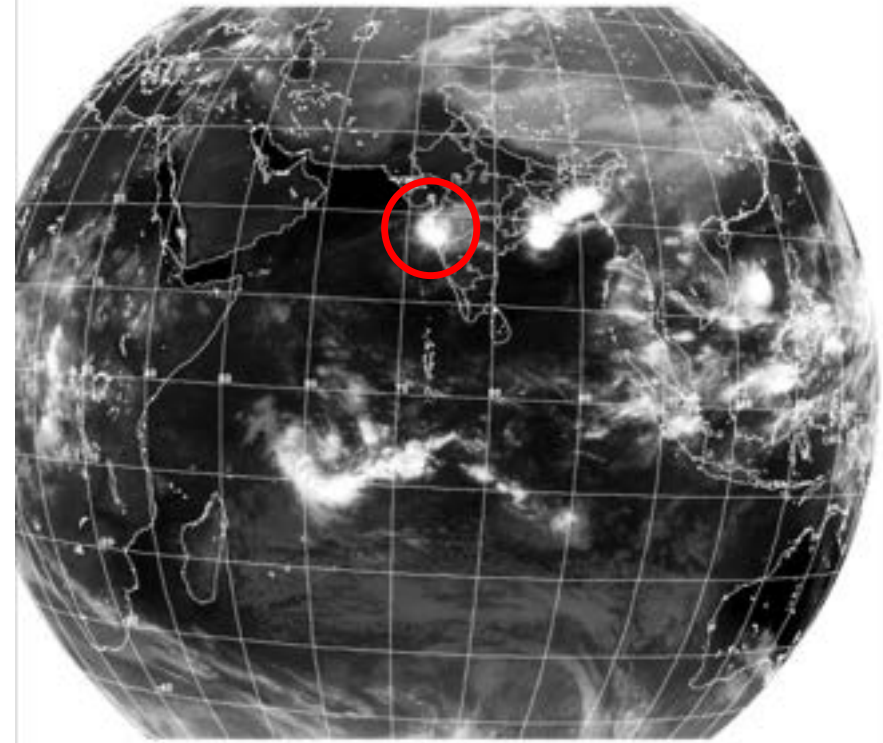
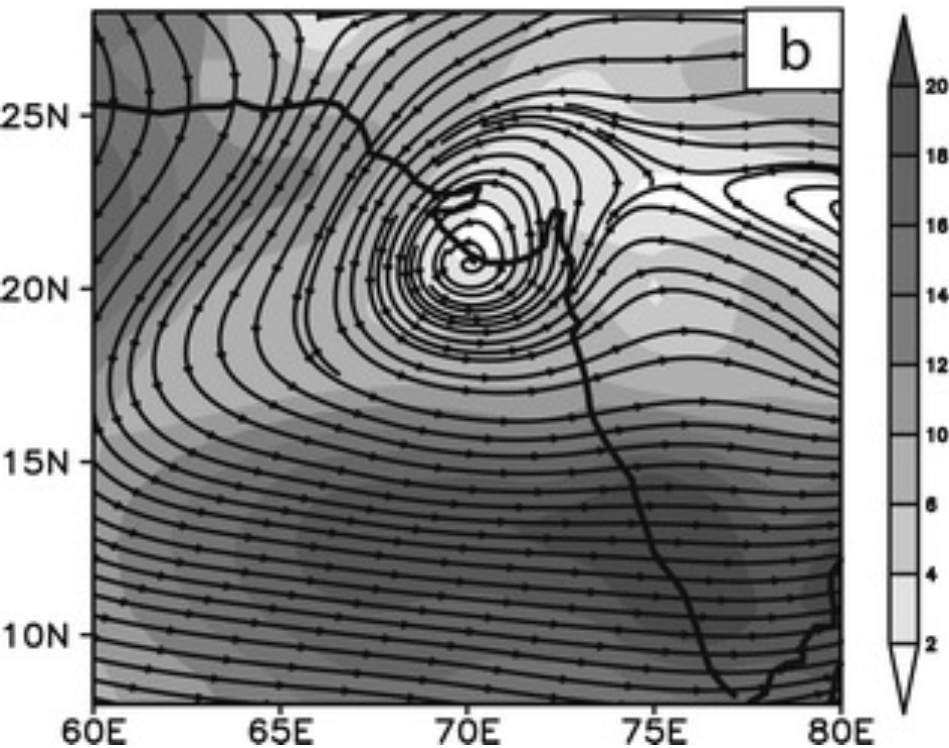




OFFSHORE VORTEX



Mid-tropospheric Cyclones (MTCs)



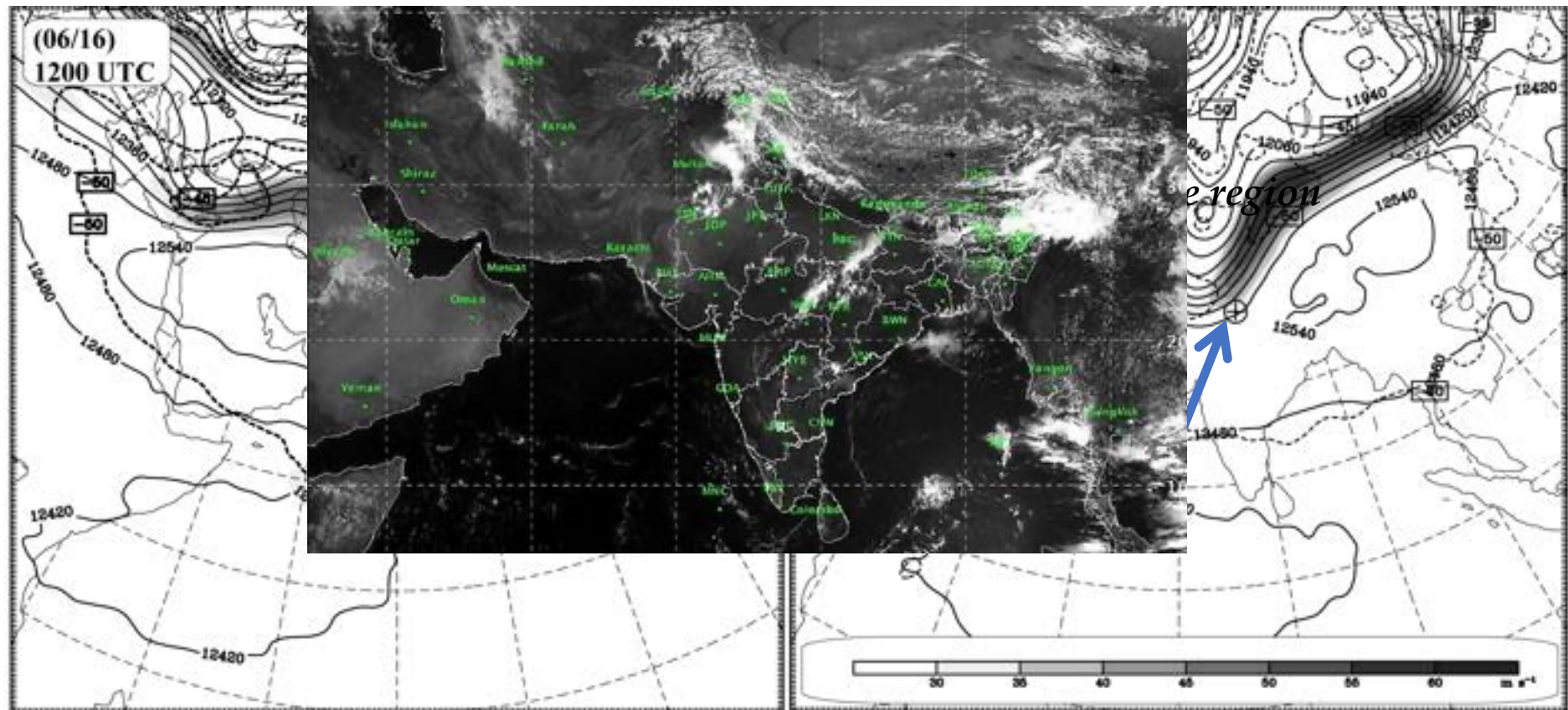
- Largest intensity is found in the middle troposphere layer (700-500 hPa) with a cold core between surface to 700 hPa and a warm core aloft.
- For initiation due to barotropic instability of the monsoon flow at 700 hPa & growth is due to latent heat.

Mid-Tropospheric Cyclones (MTCs)

- Mid-tropospheric cyclones are an important part of the monsoon circulation.
- These synoptic-scale disturbances appear with greatest intensity at levels near 500 hPa.
- It is usually characterized by very intense convective and non-convective rainfall.
- It forms mainly over the north-eastern Arabian Sea off the north Konkan-south Gujarat coast of India.
- MTCs often generated within the monsoon cyclonic shear zone at 700-500 hPa .
- They have a life span about 3-7 days and procedure very heavy rainfall over the north Konkan- Gujarat coast of India.
- Horizontal scale of 3000 km
- These are stationary systems, rarely move westward from its position
- Heavy rains occur over the Gujarat, Mumbai and Konkan:-reaches up to 20 cm per day.

Western Disturbances

- Western disturbances associated with eastward moving mid-tropospheric troughs in subtropical westerlies.
- Southward penetration of these systems cause heavy rainfall in the northern parts of India/Western Himalayas.

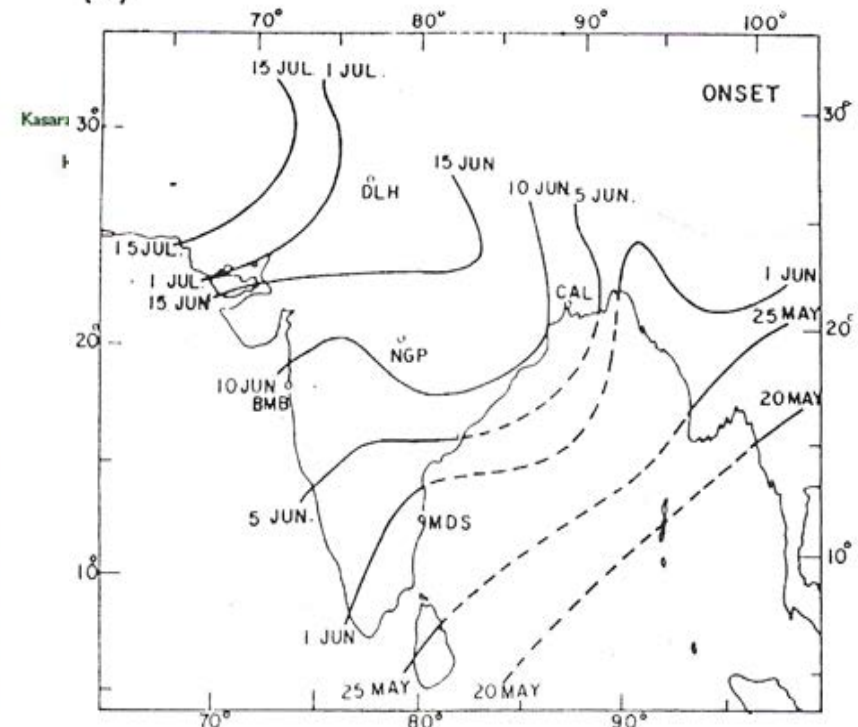


A) Onset of monsoon over Kerala

1. Rainfall Criteria

- To declare onset of monsoon over Kerala, rainfall amount over the 14 selected stations: Minicoy, Amini, Thiruvananthapuram, Panalur, Kollam, Alapuxha, Kottayam, Kochi, Trissur, Kozikode, Talassery, Cannur, Kasargode and Manglor is to be considered.
- If after *10th May*, 60% of the above 14 stations report rainfall of ≥ 2.5 mm for two consecutive days, the onset of monsoon over Kerala be declared on 2nd day provided the **WIND FIELD** and **OLR** criteria are also in concurrence.

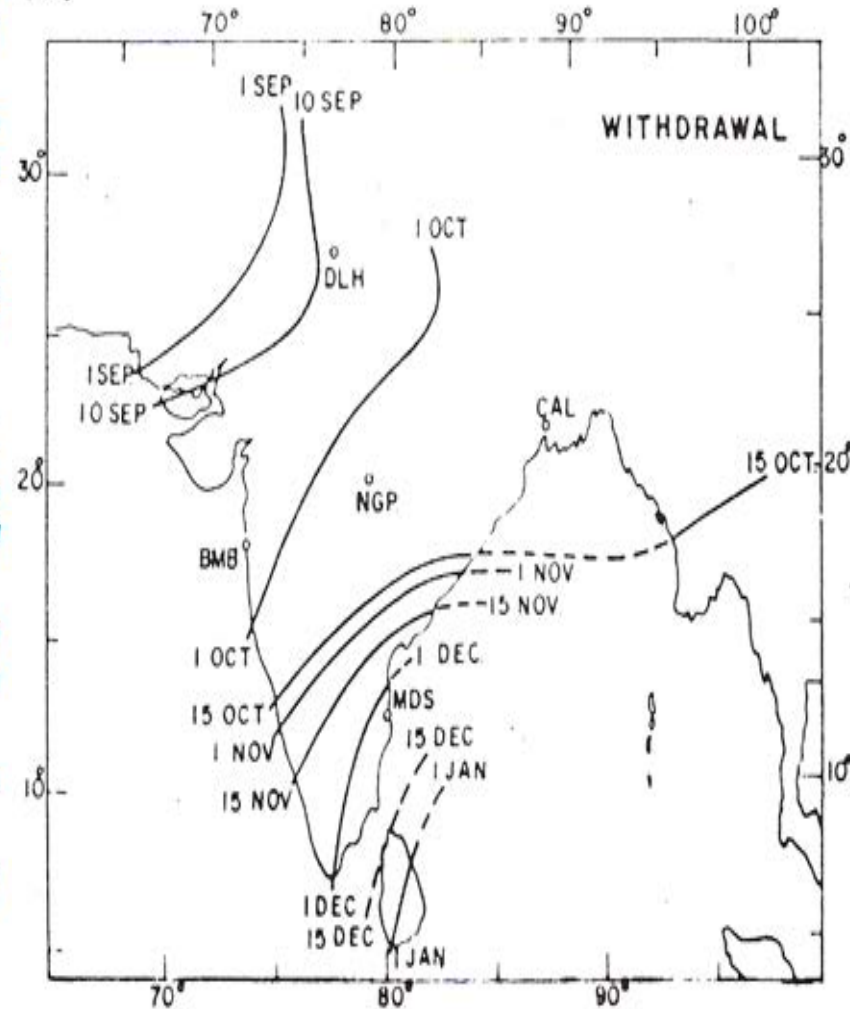
(a)



ONSET June 1; Kerala

Withdrawal of Monsoon

- The withdrawal of monsoon commences from western parts of west Rajasthan by 1st Sept.
- It withdraws from NW India by Sept 15 and the withdrawal is complete from the country outside Tamil Nadu and Kerala by Oct 15.
- By 15th October, the circulation pattern indicates a change over from the southwesterly wind regime and NE monsoon normally sets in over Tamil Nadu.
- Withdrawal is very rapid process over north India. It takes just 15 days to withdraw from NW-India; **by 1st October the NLM passes through Konkan, North Maharashtra, West M.P. and west U.P.**
- **By 15th October NML is confined to the south Peninsula and NE monsoon normally sets in over Tamil Nadu.**
- Over south Peninsula the withdrawal process is very slow.
- **By December, monsoon is completely withdrawn from India.**

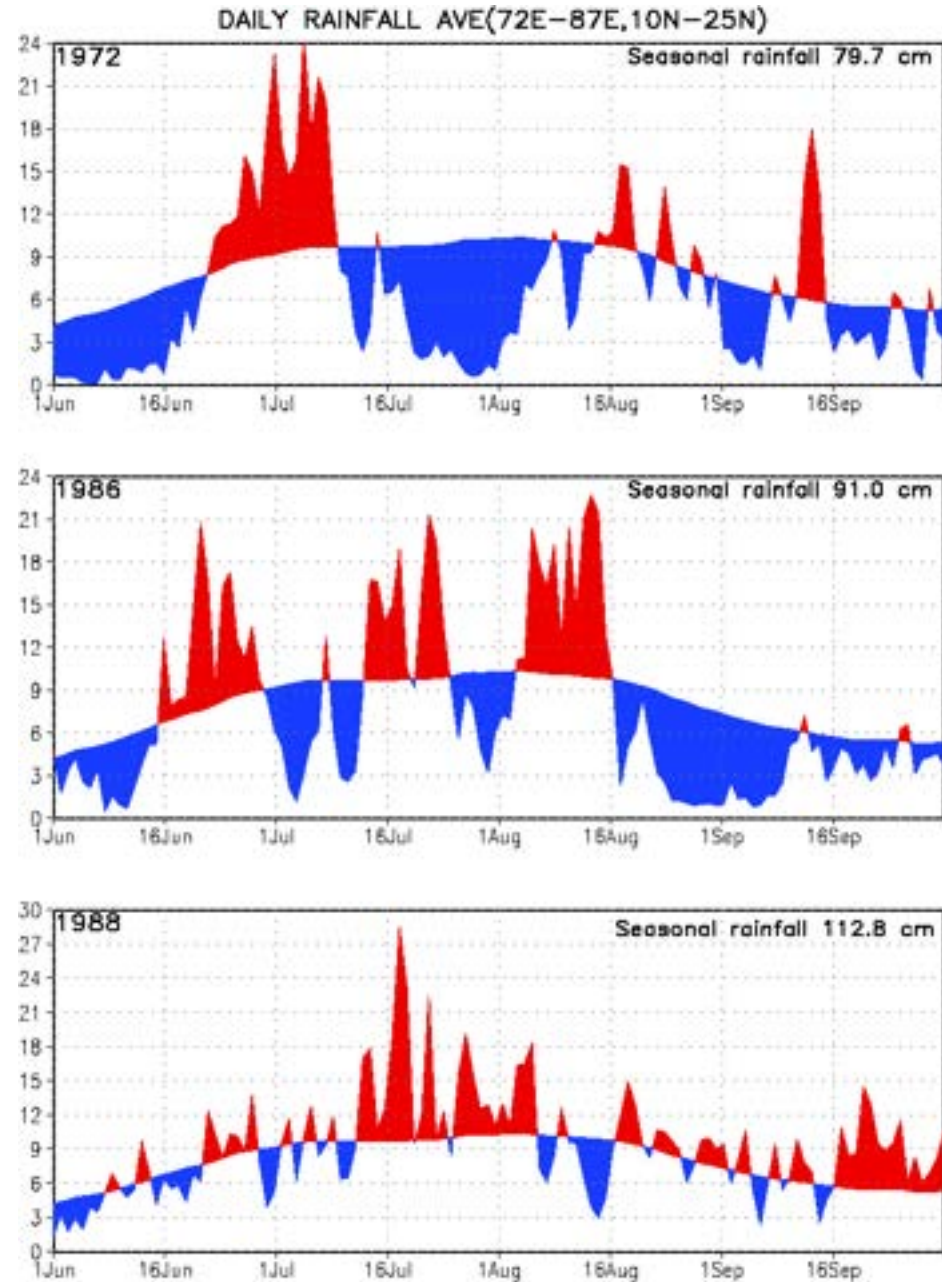


Active and Break Cycles of the Monsoon

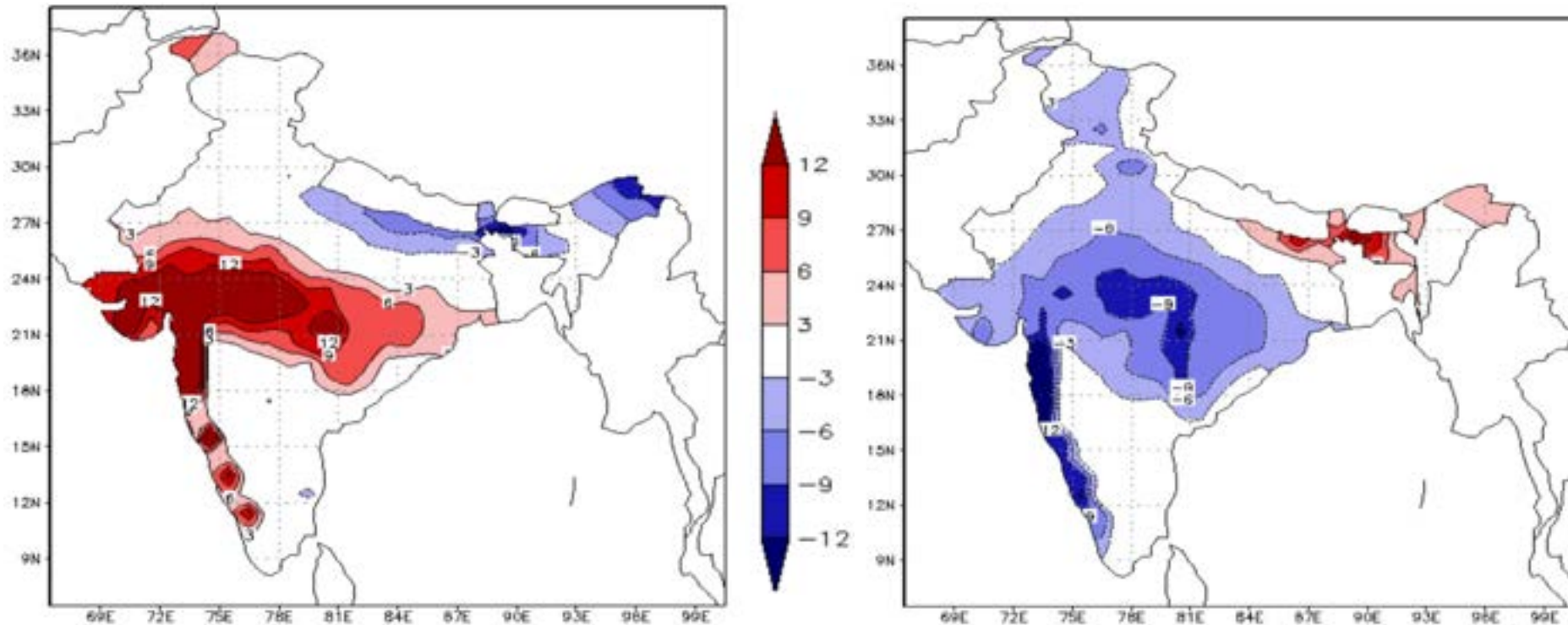
Daily rainfall (mm/day) over central India for three years, 1972, 1986 and 1988

The smooth curve shows long term mean.

Red shows above normal or wet spells while **blue** shows below normal or dry spells



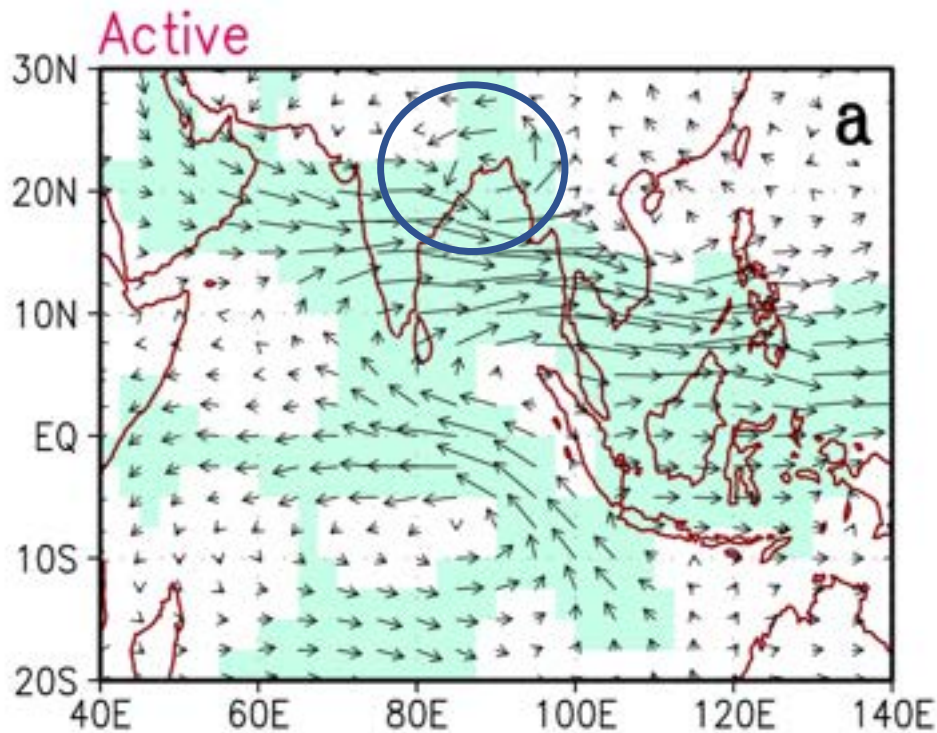
Active and Break Cycles - Rainfall



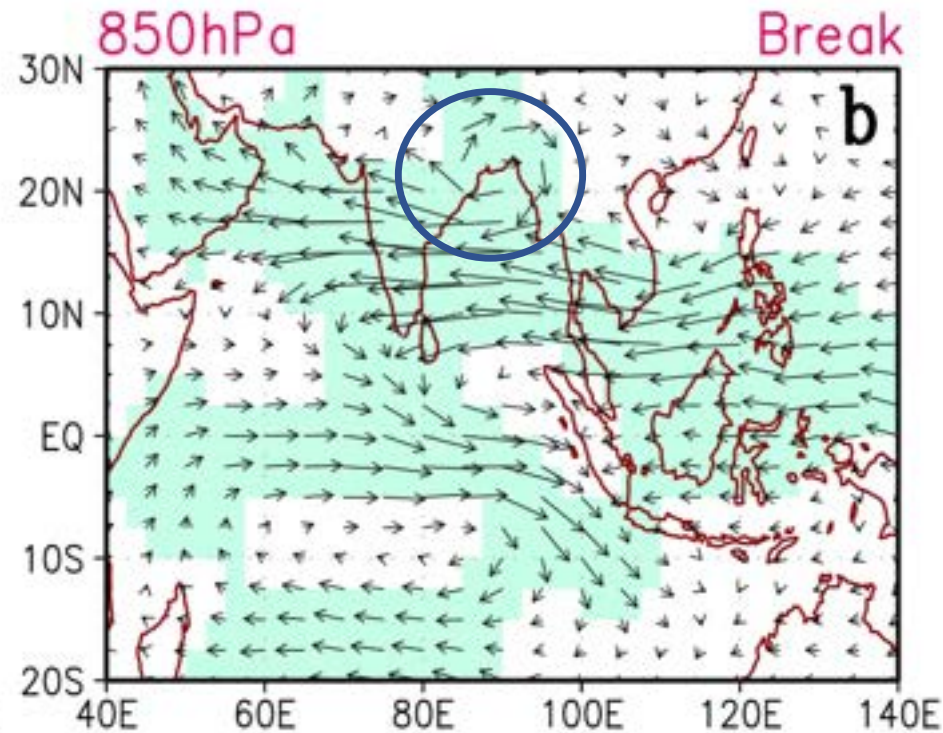
- There is more rain over central India and Western Ghats
- Less rains over the foothills of Himalayas

- There is less rain over central India and Western Ghats
- More rain over the foothills of Himalayas

Active and Break Cycles – Low level Winds



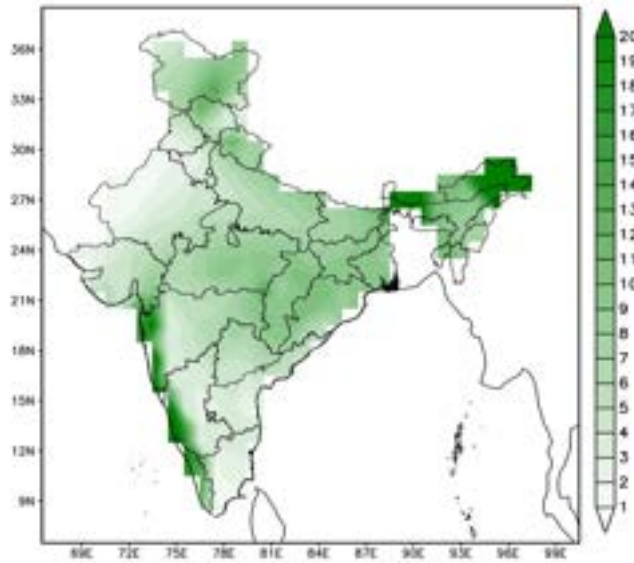
- ❖ Stronger Westerly zonal winds
- ❖ Low-level jet is strengthened
- ❖ Stronger monsoon trough – shifts south of its normal position
- ❖ Cyclonic circulation over Central-East India



- ❖ Westerly zonal winds gets Weaken
- ❖ Weak Low-level jet
- ❖ Weaker monsoon trough- shifts north of its normal position
- ❖ Anti-cyclonic circulation over Central-East India

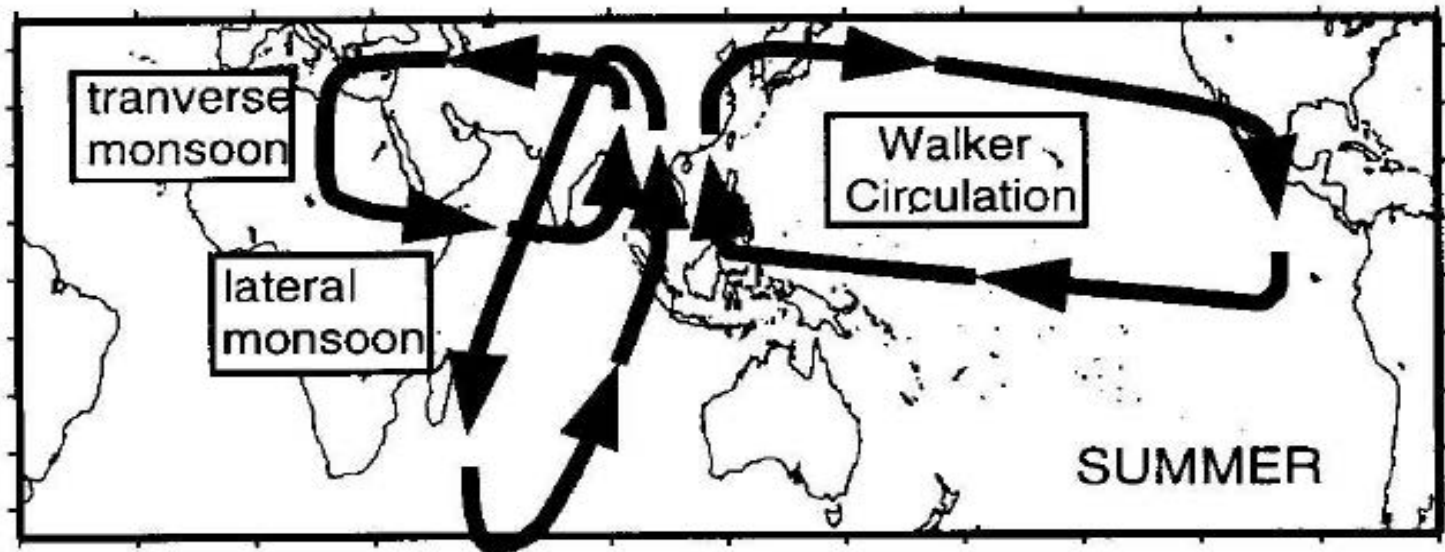
These spells occur at 30–60 day scales - Monsoon Intraseasonal Oscillations

Interannual Variations of the Indian Monsoon



Walker and Hadley circulations are an important circulation components of the monsoon

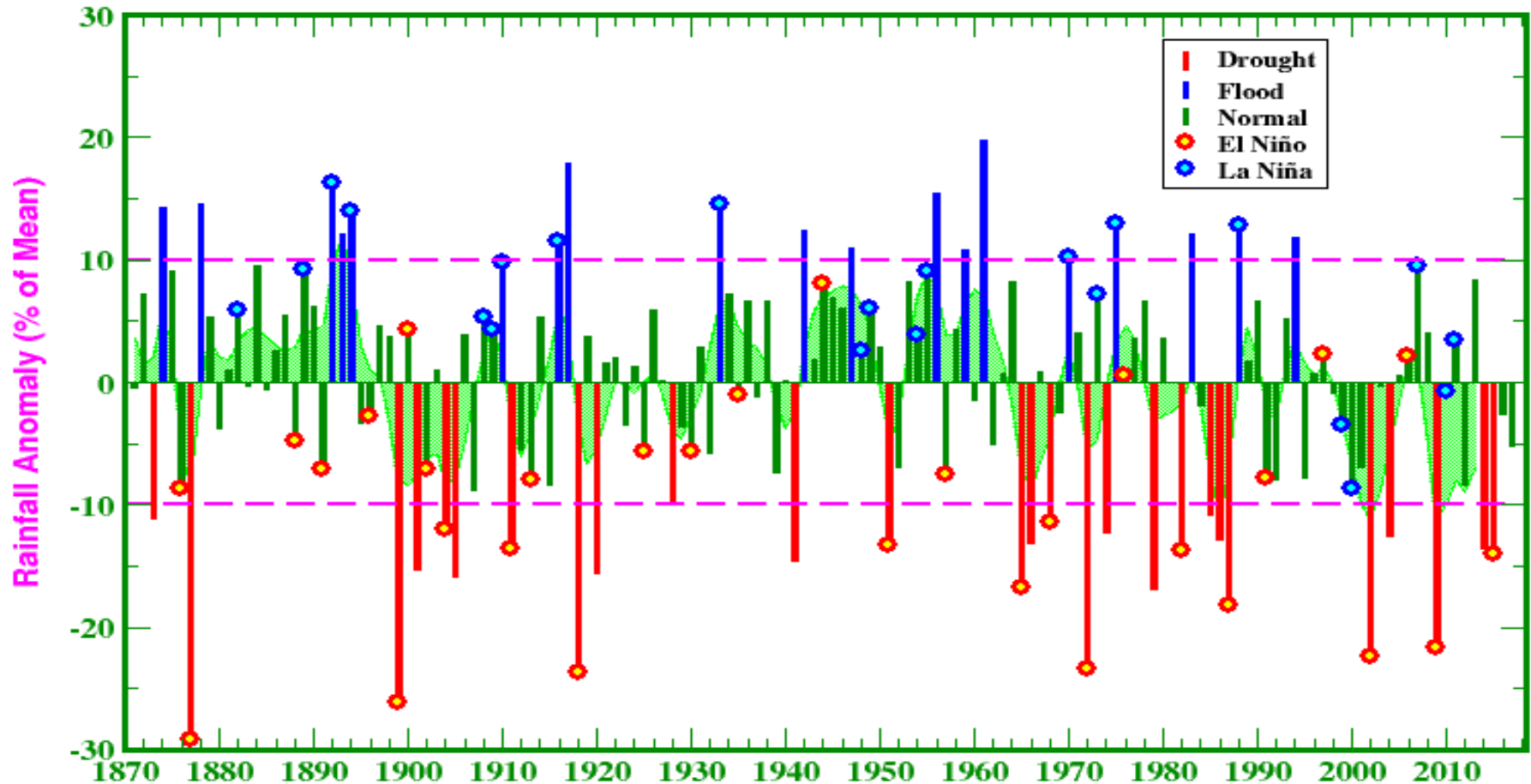
Year to Year variations of the Indian monsoon is characterized by fluctuations of these cells



Interannual Variations of the Indian Monsoon

All-India Summer Monsoon Rainfall, 1871-2017

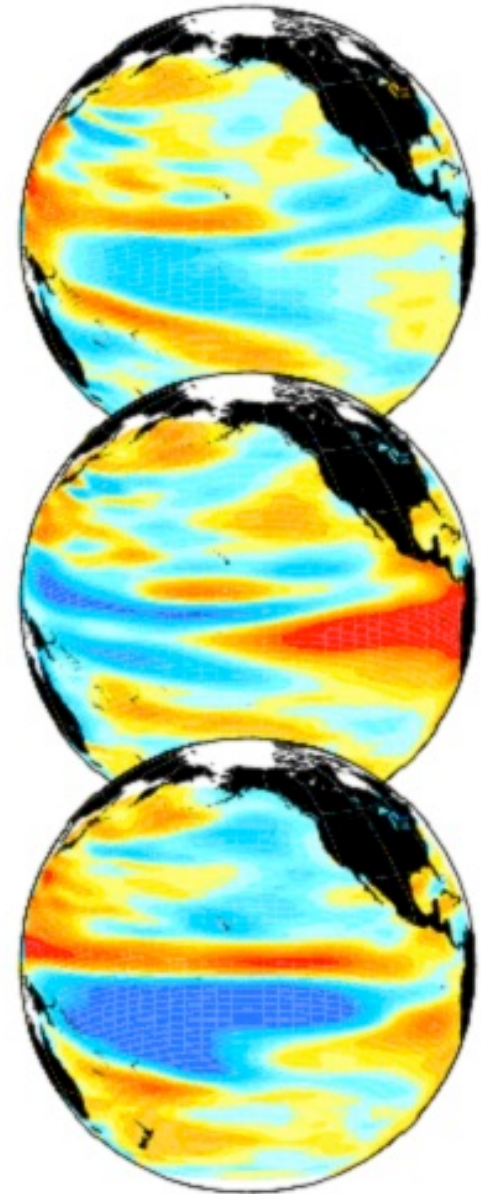
(Based on IITM Homogeneous Indian Monthly Rainfall Data Set)



ENSO is known to be the most important “climate driver” of the Indian summer monsoon

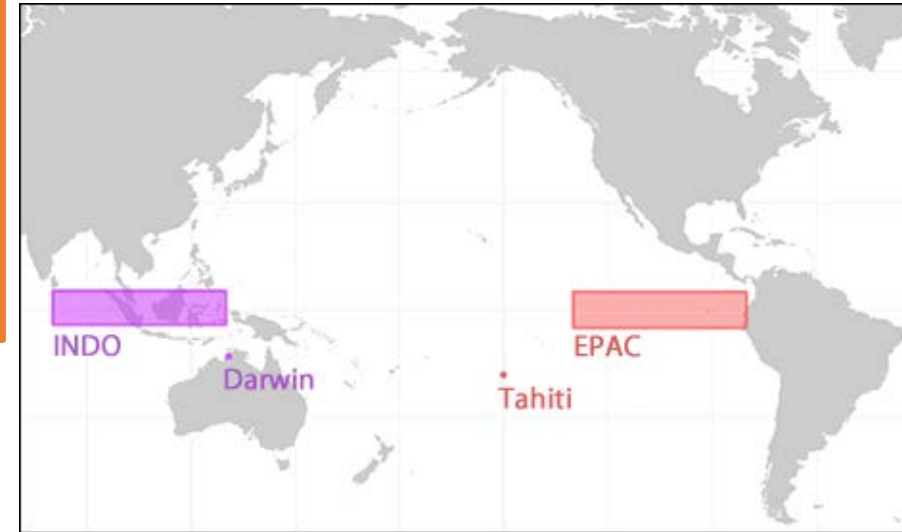
El Niño (EN) & Southern Oscillation (SO)- ENSO

- "ENSO" refers to the El Niño/Southern Oscillation, the interaction between the atmosphere and ocean in the tropical Pacific
- It is the strongest natural inter-annual climate signal, which has widespread effects on the global climate system.
- It occurs at irregular intervals of 2-7 years and usually lasted 1-2 years.
- El Niño: A warming of the ocean surface, or above-average sea surface temperatures (SST), in the central and eastern tropical Pacific Ocean.
- La Niña: A cooling of the ocean surface, or below-average sea surface temperatures (SST), in the central and eastern tropical Pacific Ocean. The normal easterly winds along the equator become even stronger.
- Normal: Often tropical Pacific SSTs are generally close to average.



The Southern Oscillation (SOI)

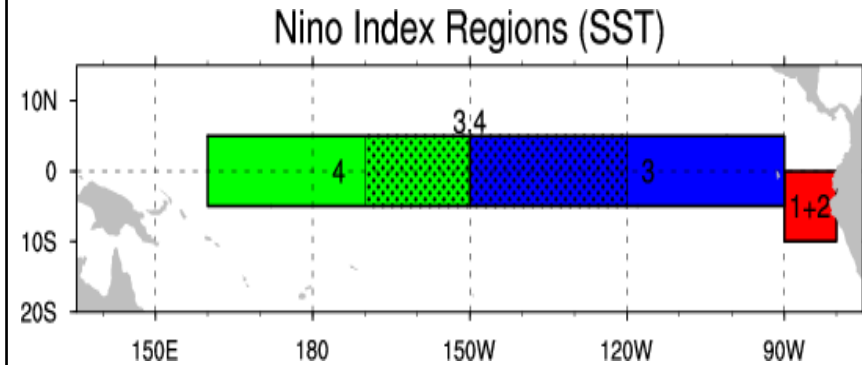
- The Southern Oscillation is the atmospheric component of El Niño.
- Naturally occurring cyclical change in wind circulation/Pressure patterns
- The SOI is computed from fluctuations in the surface air pressure difference between Tahiti and Darwin



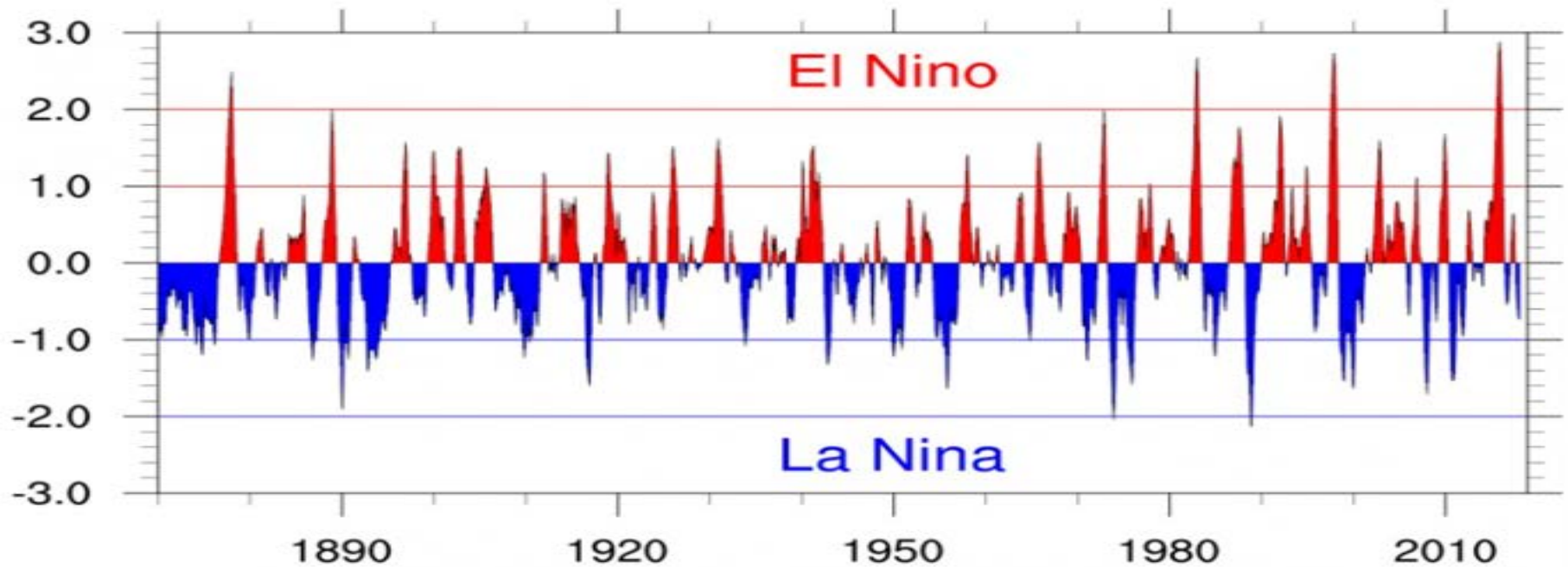
- ❖ El Niño episodes have negative SOI, meaning there is lower pressure over Tahiti and higher pressure in Darwin.
- ❖ La Niña episodes have positive SOI, meaning there is higher pressure in Tahiti and lower in Darwin.

How to Monitor ?

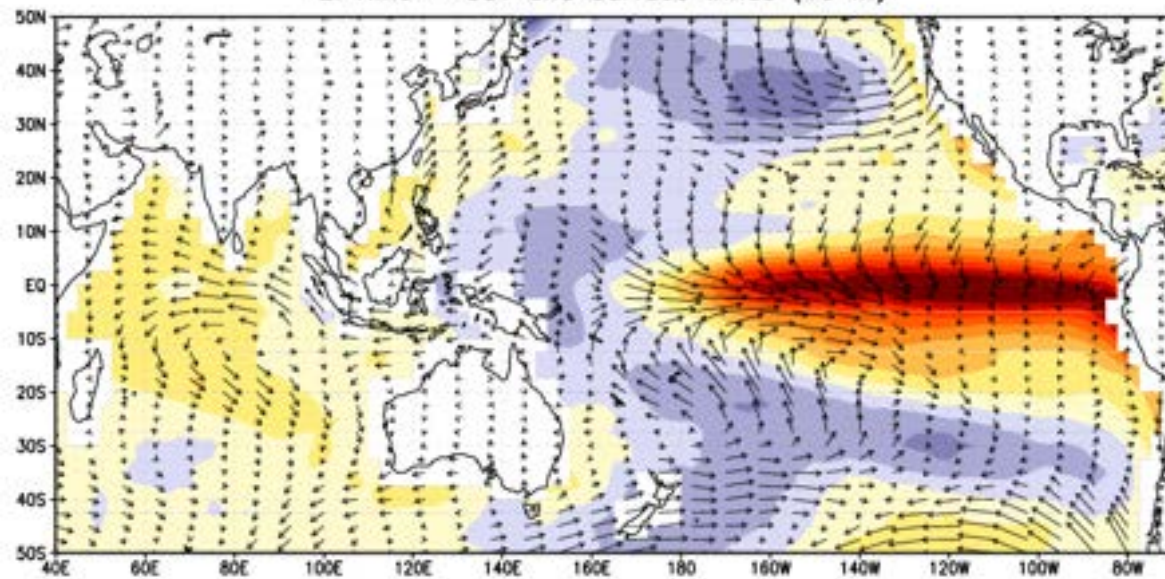
- The Oceanic Nino Index (ONI) is one of the primary indices used to monitor the El Nino-Southern Oscillation (ENSO).
- The ONI is calculated by averaging sea surface temperature anomalies in an area of the east-central equatorial Pacific Ocean, which is called the **Nino-3.4** region (5S to 5N; 170W to 120W).



ONI: 1870-2018: Reference: 1960-1989



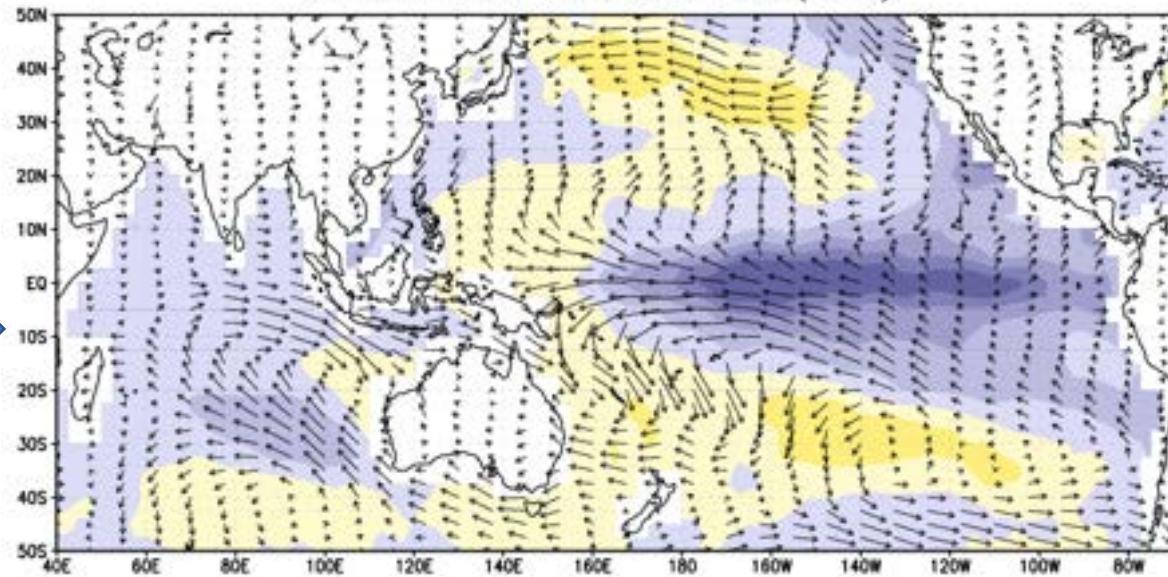
El Nino - SST and Surface winds (10 m)



← El Nino

La Nina - SST and Surface winds (10 m)

La Nina →



Normal Conditions

West Pacific:

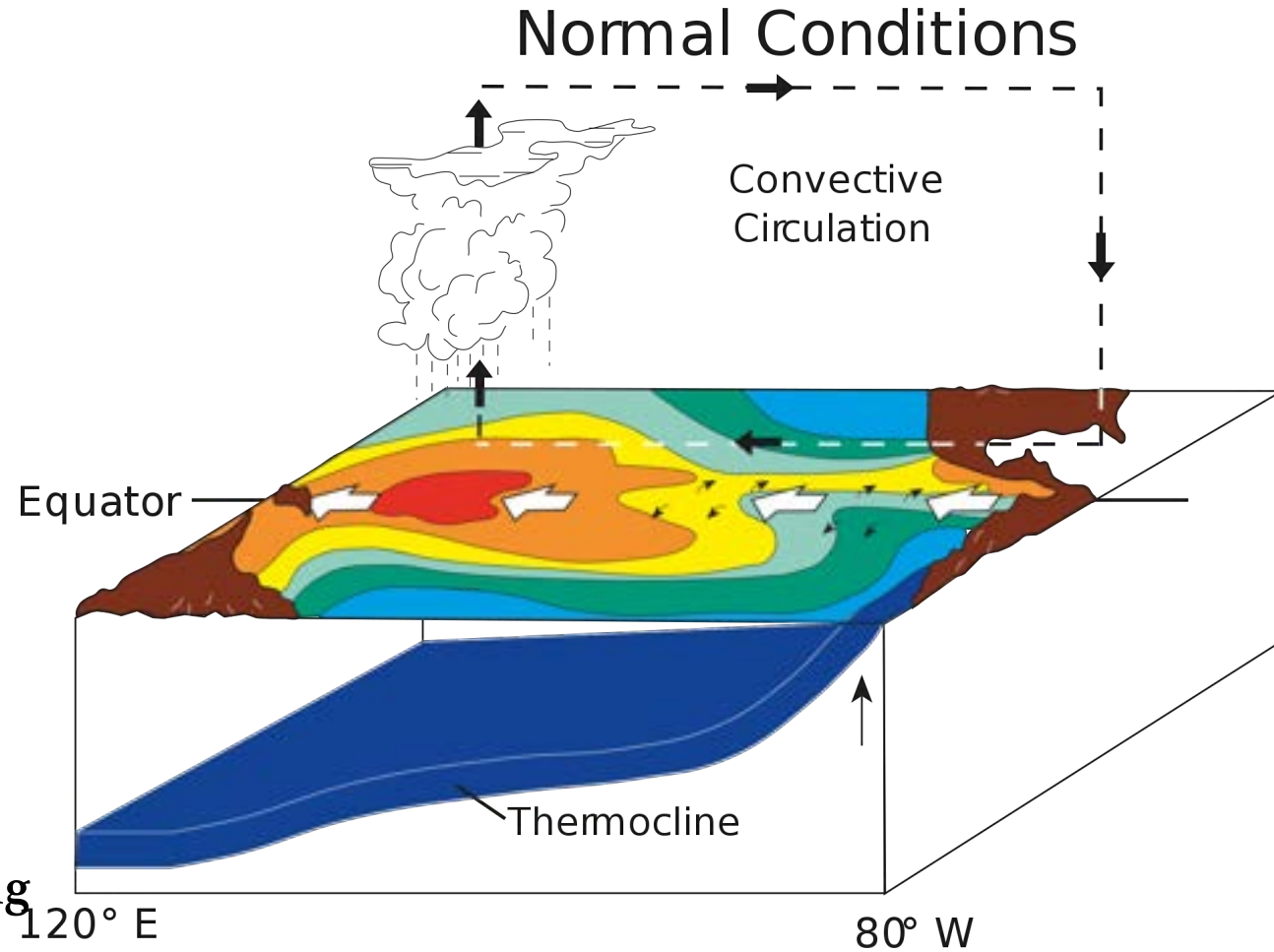
Warm ocean water
Low pressure

Unstable
atmospheric
conditions

East Pacific:

Cold water upwelling
High pressure

Stable atmospheric
conditions



El Nino Conditions

Tradewinds slow and reverse direction

Eastern Pacific

Upwelling ceases

Warmer surface water

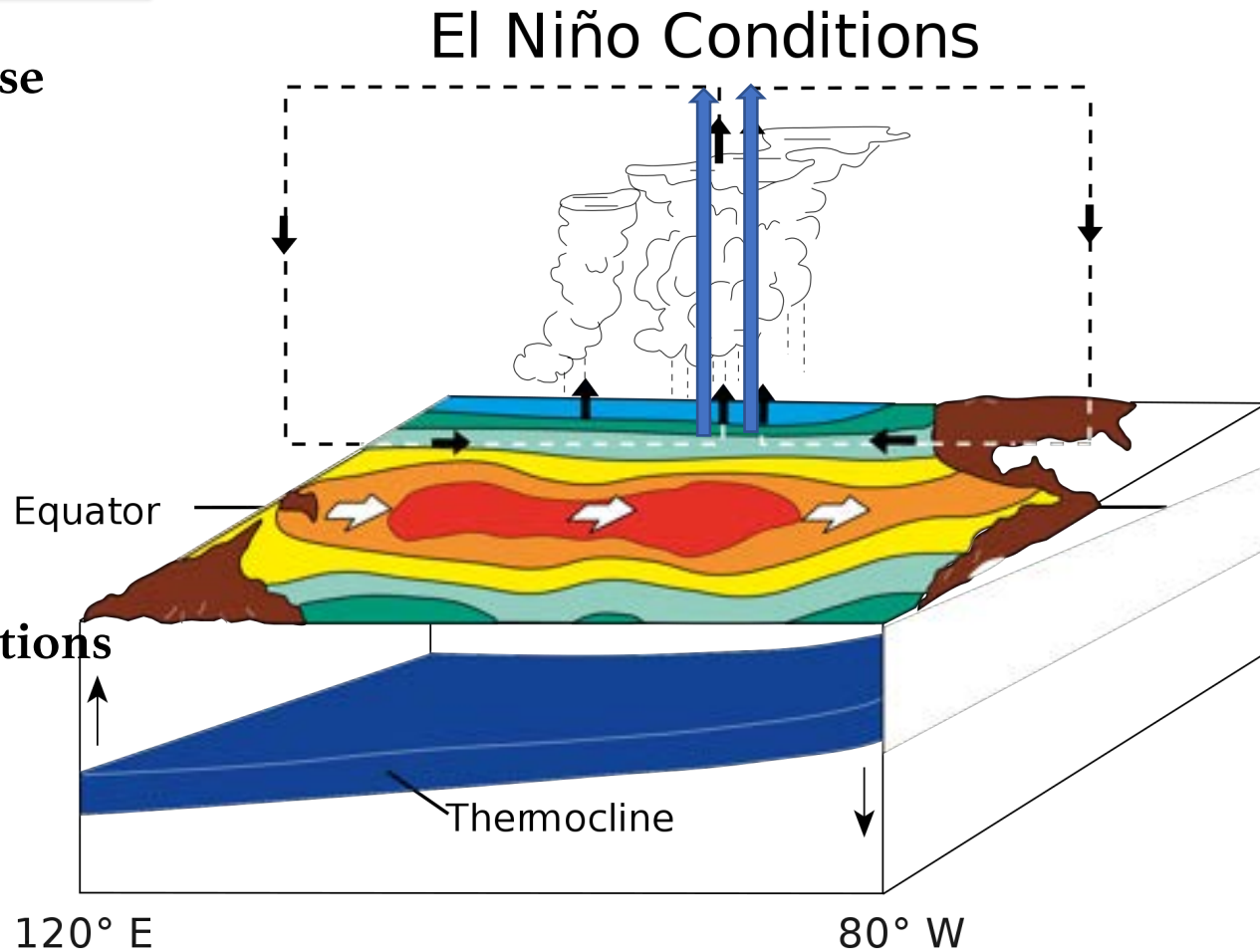
Wet and unstable conditions

Western Pacific

Cooler surface water

Low pressure weakens

Drier and stable conditions



La Nina Conditions

Amplification of normal conditions

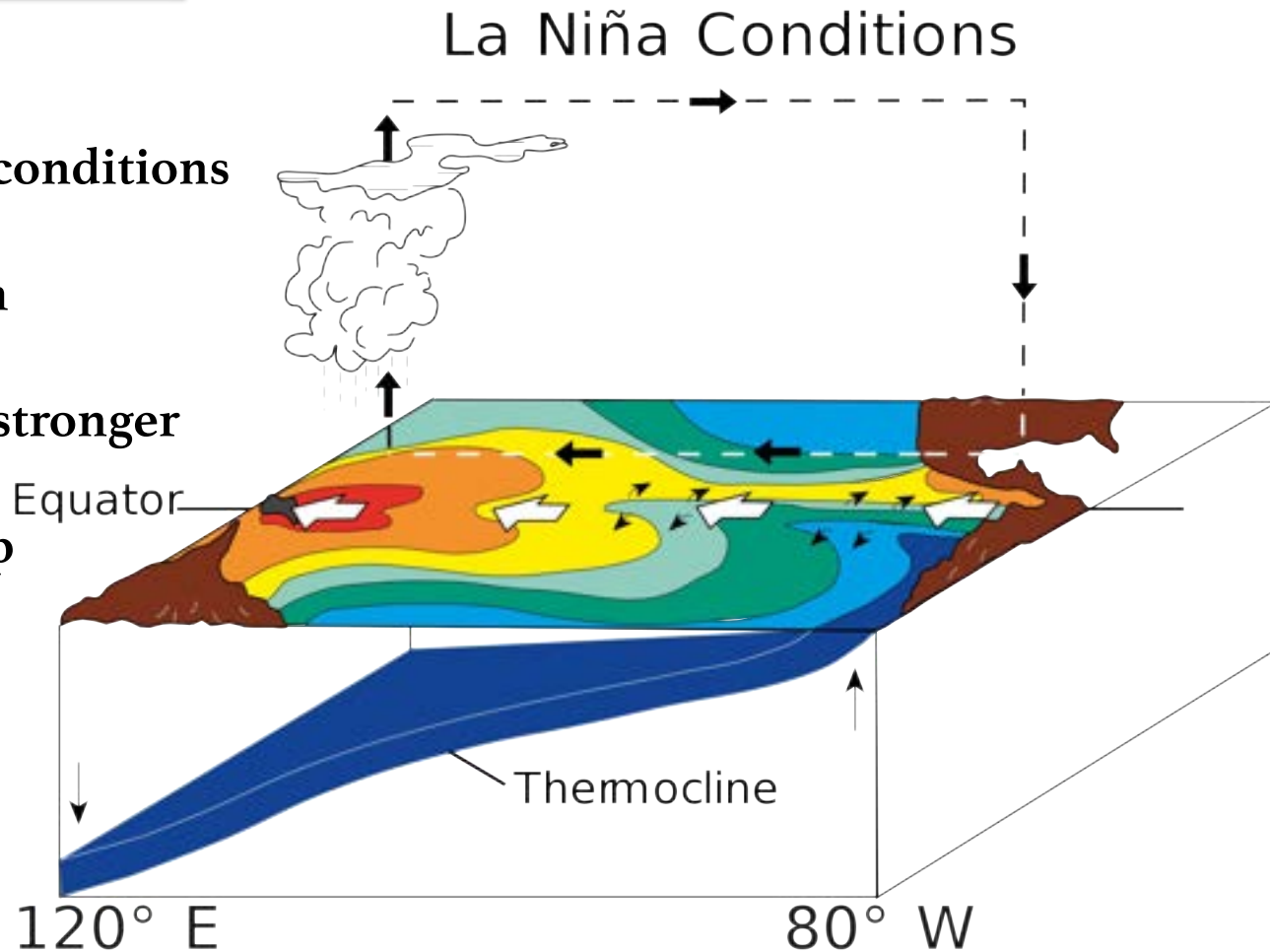
Strong Walker circulation

Easterly trade winds are stronger

More warm water piles up
in western Pacific

Deep Low pressure

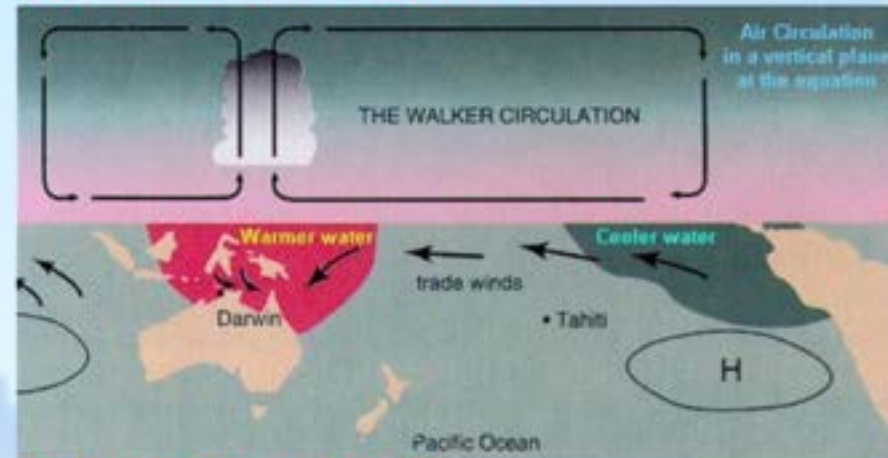
More wet and unstable
in western Pacific



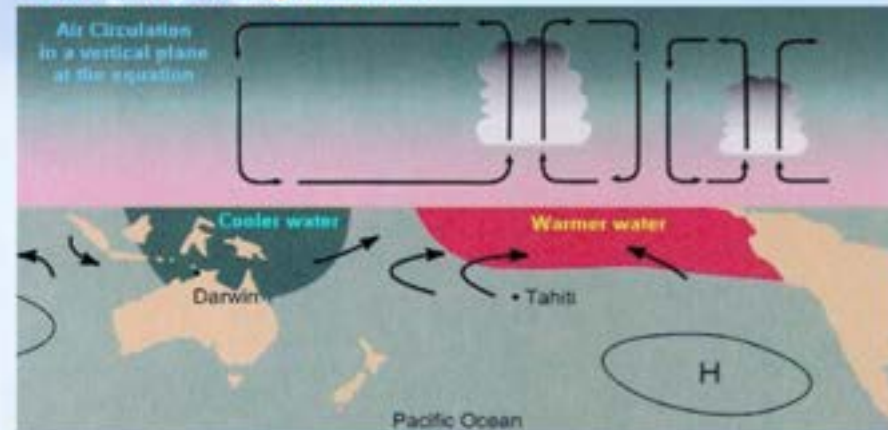
ENSO-Dynamics

- ❖ The prevailing easterly trades converge over Indonesia in conjunction with the Asian monsoon, producing widespread convection. Additionally, warm water "piles up" in the western Pacific, due to the easterly winds. Further east, the SE trades and equatorial easterlies in the eastern and central Pacific produce upwelling of cool water along the equator and coast of South America.
- ❖ During El Niño, the easterlies relax, reducing the amount of upwelling and allowing the western warm water to move eastward. As time goes on, the warm pool in the western Pacific grows and expands eastward toward the central Pacific. Once the warmest water reaches the International Date Line, anomalous convection appears in that region, accompanied by a weakening of the equatorial easterlies.
- ❖ This pattern typically may be preceded or followed by a warming that causes the ITCZ to move farther south than normal, which contributes to enhanced rainfall across Ecuador and northern Peru, producing the "years of abundance (periods of extreme wetness along the normally very dry Peruvian coast)".

Typical Walker Circulation System



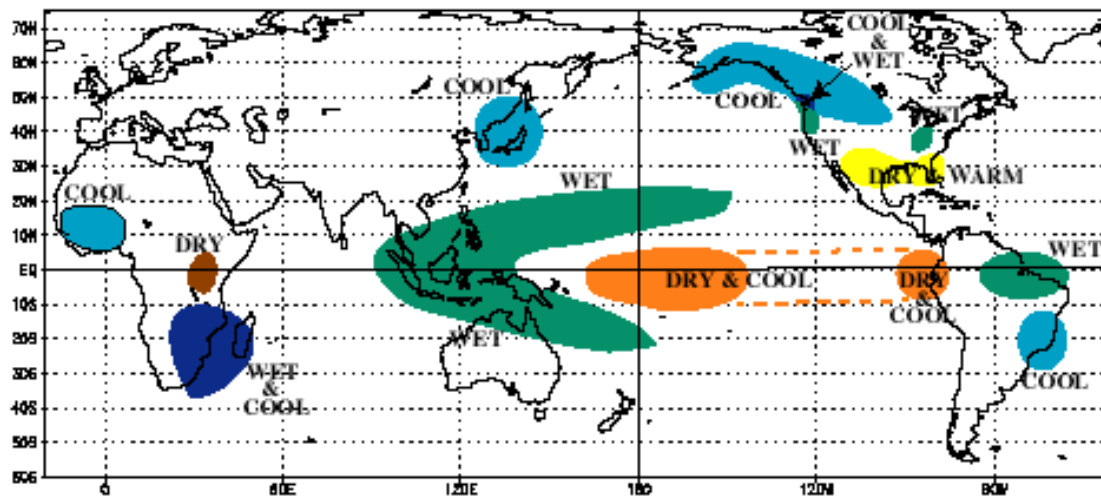
Circulation during an El Niño



- The typical wind and water circulation pattern.
- The circulation pattern during an El Niño event.

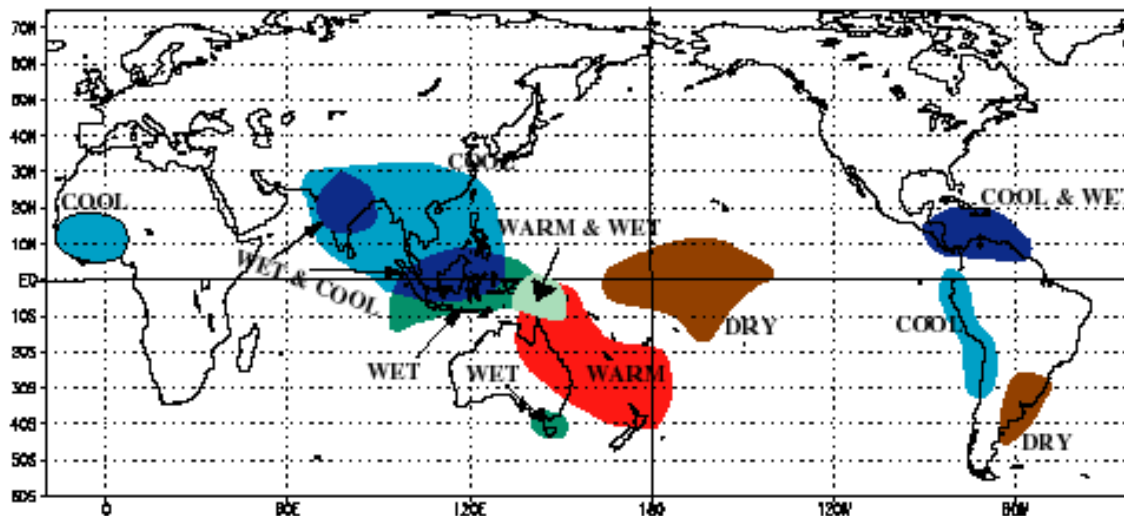
ENSO impacts on Global Climate

COLD EPISODE RELATIONSHIPS DECEMBER - FEBRUARY



Opposite patterns prevail during Warm Phase of ENSO

COLD EPISODE RELATIONSHIPS JUNE - AUGUST

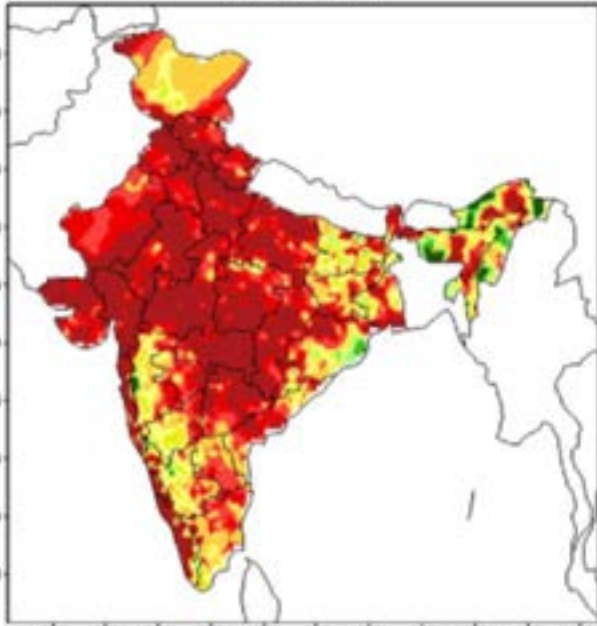


ENSO-Key points

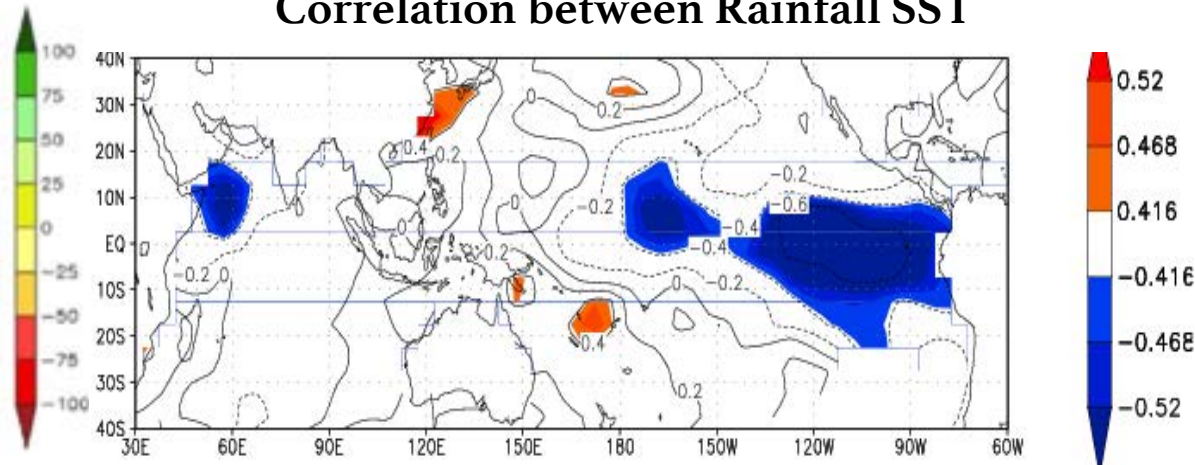
- El Nino is a phenomenon of anomalous warming/weak trade winds of the central and east equatorial Pacific Ocean with a quasi periodicity of 2-7 years.
- La Nina is just the opposite phase of El Nino and is associated with anomalous cooling/strong trade winds over the equatorial Pacific Ocean.
- They tend to develop during the period April- June and reach their maximum strength during December -February.
- Recur every 2 to 7 years
- Southern Oscillation the atmospheric component of El Nino
- It is an oscillation in surface air pressure between the tropical eastern and the western Pacific Ocean waters.
- The strength of SO is measured by the Southern Oscillation Index (SOI) defined as the normalized difference in surface pressure between Tahiti, and Darwin (Tahiti minus Darwin)
- High SOI (large pressure difference) is associated with stronger than normal trade winds and La Niña conditions, and low SOI (smaller pressure difference) is associated with weaker than normal trade winds and El Niño conditions.

ENSO-Monsoon Relationship: Teleconnection

Rainfall during El Nino Years



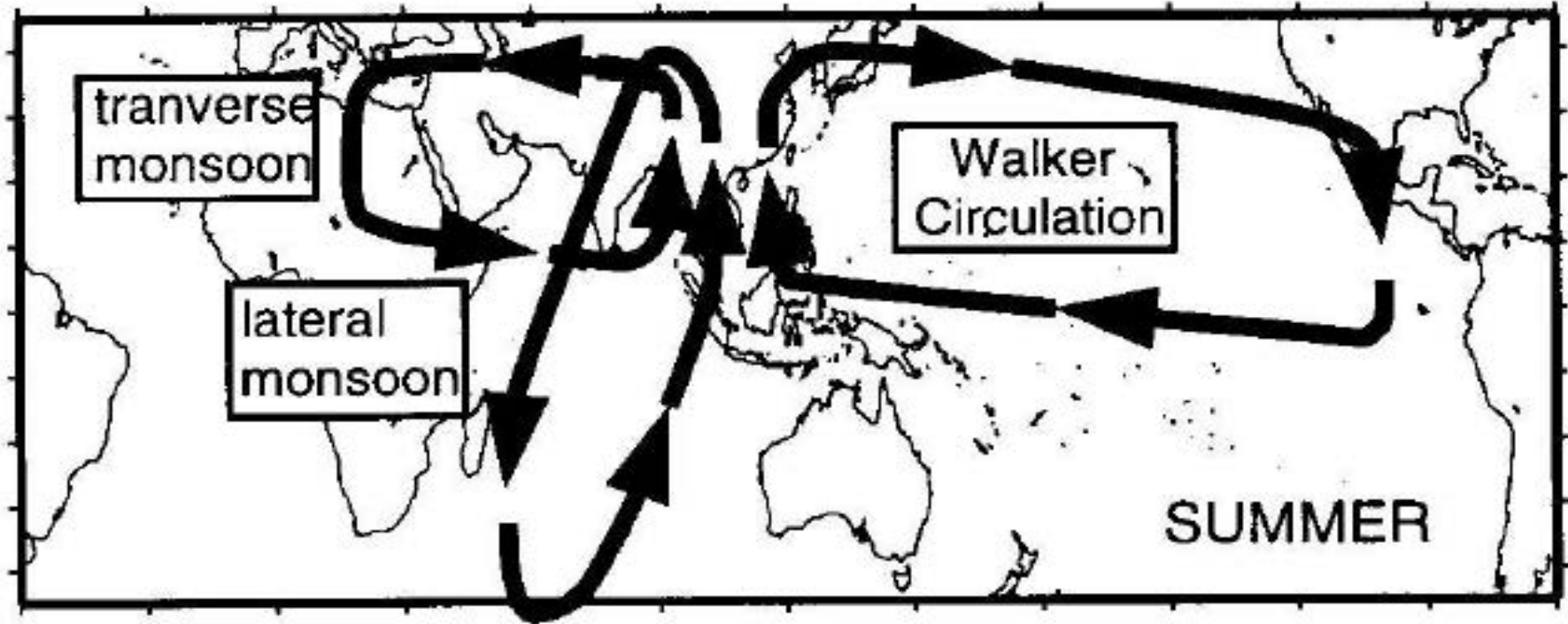
Correlation between Rainfall SST



During El Niño years, in general, most parts of the country except northeast India receive below normal rainfall anomalies. The impact is more over northwest India and central India.

The negative correlation between Indian monsoon and ENSO is the strongest for east Pacific sea surface temperature)

ENSO-Monsoon Relationship: Mechanism



Basic Mechanism lies in interaction between Monsoon Hadley Cell and Walker Cell

ENSO-Monsoon Relationship: Mechanism

- ❖ The inter-annual variation of the Indian monsoon is characterized by fluctuations of a regional Hadley circulation.
- ❖ A strong monsoon is associated with anomalous ascent around 25N and a weak monsoon is associated with anomalous ascent near the equator.
- ❖ ENSO influences the Indian monsoon through the interaction between the equatorial Walker circulation and the regional monsoon Hadley circulation
- ❖ The decrease in the Indian monsoon rainfall associated with the warm phases of ENSO is due to an anomalous regional Hadley circulation with descending motion over the Indian continent and ascending motion near the equator sustained by the ascending phase of the anomalous Walker circulation in the equatorial Indian Ocean